

Unit 4 Overview: The Coordinate Plane

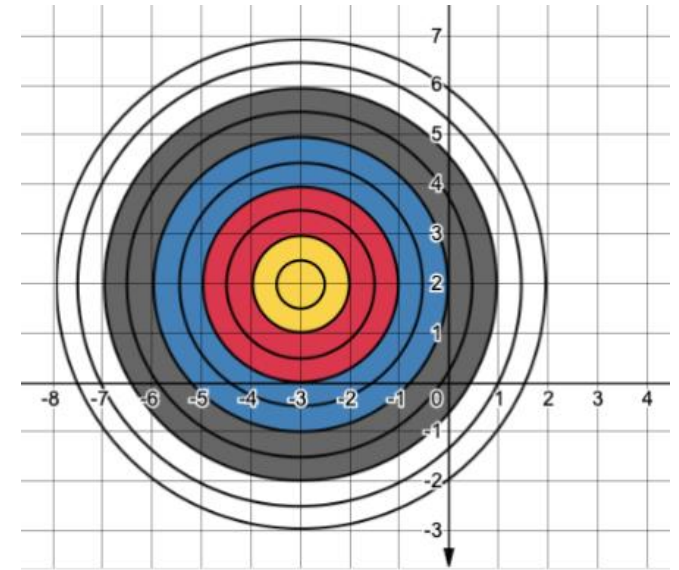
Unit Narrative



Students were introduced to the coordinate plane in Grade 5 mathematics as they made connections between the number line and the coordinate plane. Instruction in Grade 5 focused only on quadrant 1.

In Grade 6 Accelerated, students extend their understanding of the coordinate plane to include all four quadrants. Students review plotting integers and comparing their locations on the coordinate plane in relation to the axes, the different quadrants, and their relationships to each other. Students use this knowledge to help them compare and explore the similarities and differences between the coordinates of points that make a vertical or horizontal line. Students then learn about reflecting points on the coordinate plane over the x - and y -axis. Instruction here lays the foundation for transformations they will perform on figures in the coordinate plane in the Grade 7 Accelerated Math course.

Students extend their knowledge of plotting points in Section 2 to include finding the distance between points. Students discover the process to finding distance between points on the coordinate plane when coordinates are on the same or different sides of the x - or y -axis. Students apply their knowledge of absolute value to write equations for finding distance on the coordinate plane. Finally, students are able to build on this learning and use distance on the coordinate plane to find side lengths of rectangles, in turn being able to find area and perimeter on the coordinate plane. Instruction here lays the foundation for finding the diagonal distance between two points using the Pythagorean Theorem in Grade 7 Accelerated Math.



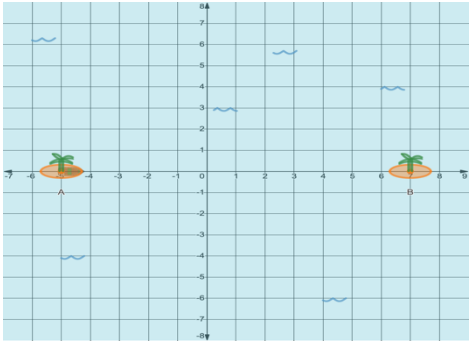
Unit At-A-Glance

Section 1: Plotting Points on the Coordinate Plane	Benchmarks Addressed	Lesson 1	Plotting Integers Using One-Unit Scales for Axes
	MA.6.GR.1.1	Lesson 2	Points on the Coordinate Plane – Part 1
		Lesson 3	Points on the Coordinate Plane – Part 2
Section 2: Distance on the Coordinate Plane	Benchmarks Addressed	Lesson 4	Distance on the Coordinate Plane
	MA.6.GR.1.2 MA.6.GR.1.3	Lesson 5	Perimeter on the Coordinate Plane
		Lesson 6	Solving Problems Involving Rectangles on the Coordinate Plane



Differentiation

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Engagement <i>The “WHY” of learning</i>	In Lessons 2 and 3, students plot rational numbers on the coordinate plane. To provide additional entry points, consider substituting some of the rational numbers with simpler fractions and decimals. Students who may struggle with accuracy plotting on the coordinate plane may benefit from practicing with familiar fractions and decimals (such as $\frac{1}{2}$) before moving on to plotting more complex rational numbers.
Representation <i>The “WHAT” of learning</i>	- Utilize Desmos graphs as an alternate way for students to graph points or shapes on a coordinate plane. - Students can use the Desmos graphing calculator to verify their responses as means for self-regulation. - Students can explore values and ordered pairs on a coordinate plane in real time by interacting digitally with the graphing calculator. For example, students can follow the graph of a line with their mouse to make sense of ordered pairs and how those ordered pairs make up that line.
Action & Expression <i>The “HOW” of learning</i>	Transform your classroom into a life-sized coordinate plane, with your students acting as points on the coordinate plane. Students can then travel from one point to another and count out the distance between those points. 

ELL Information

Desmos In Languages: Mini Golf Marbleslides

Consider incorporating Desmos activity “Mini Golf Marbleslides” for students in the development stages of language acquisition. Desmos offers this activity in a variety of languages, so students can engage with the content in their native language while developing their vocabulary with the coordinate plane.

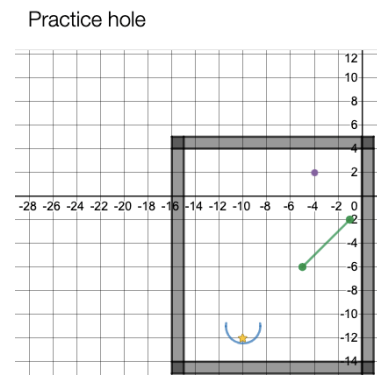
Consider including the “Mini Golf Marbleslides” Desmos activity into this unit. This activity is 30 – 45 minutes long and allows students to practice plotting points on the coordinate plane. This activity would fit nicely into Lesson 1 or Lesson 2.

- Create a class code for this activity in Desmos in your Teacher Dashboard.
- This activity is available to translate into Spanish and French.

Glossary Videos

Utilize the following glossary videos, as they are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from Additional Differentiated Support below:

- Absolute value
- Axes (of a graph)
- Coordinate
- Coordinate plane
- Origin
- Quadrant
- Scale



Additional Differentiated Support

Scaffolding Resources

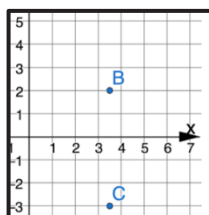
Card Sort

Create sets of matching cards, with one card having a visual of two points on the coordinate plane and the other card having the vertical or horizontal distance between those two points. Students can use these cards in a variety of ways.

- Play Memory by flipping the cards over and trying to get a match.
- Play Go Fish with a partner to find the matching points with their distance on the coordinate plane.
- Time themselves and see how fast they can find the matching cards.

Example Card Set:

Card A



Card B

Distance : 5 units

You could also add the equation used to find the distance between the two points on Card B as well, i.e., “Distance: $|2| + |-3| = 5$.”

On-Ramp

On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills:

- Coordinate Points in Real-World Context
- Horizontal Distance on the Coordinate Plane
- Vertical Distance on the Coordinate Plane
- Number Lines: Whole Number Sums and Differences
- Adding Decimals
- Subtracting Decimals
- Using the Perimeter Formula for Rectangles
- Finding Perimeter of Rectilinear Shapes with Missing Sides
- Finding Perimeter of Rectilinear Shapes
- Comparing Shapes
- Create Ordered Pairs from a Pattern



Section Overview

Section 1: Plotting Points on the Coordinate Plane (Lessons 1-3)

Benchmarks Addressed	Learning Targets	
MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x – or y - axis as the line of reflection when two ordered pairs have an opposite x – or y –coordinate.	Lesson 1	I can plot integers using one-unit scales for axes.
	Lesson 2	- I can plot integers using different scales. - I can plot rational numbers in the form of decimal and fractional ordered pairs. - I can select an appropriate scale given a set of ordered pairs
	Lesson 3	I can plot rational numbers using a scale of 1.

Section 2: Distance on the Coordinate Plane (Lessons 4-6)

Benchmarks Addressed	Learning Targets	
MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x –coordinate or the same y –coordinate, represented on the coordinate plane. MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle. <i>Clarification 1: Instruction includes finding distances between points, computing dimensions of a rectangle or determining a fourth vertex of a rectangle.</i> <i>Clarification 2: Problems involving rectangles are limited to cases where the sides are parallel to the axes.</i>	Lesson 4	I can find the distance between two ordered pairs.
	Lesson 5	- I can find distances between ordered pairs. - I can find the dimensions of a rectangle on a coordinate plane. - I can find the perimeter of a rectangle on a coordinate plane.
	Lesson 6	- I can find distances between ordered pairs. - I can find the area of a rectangle on a coordinate plane. - I can find the fourth vertex of a rectangle on a coordinate plane.



Vertical Alignment

Grade 6 Accelerated Unit 4
The Coordinate Plane



Grade 7 Accelerated Unit 6
Relationships in Triangles

Prior Course(s)	Course Level	Future Course(s)
MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane. MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.	MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- or y-coordinate. MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x-coordinate or the same y-coordinate, represented on the coordinate plane. MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.	MA.8.GR.1.1 Apply the Pythagorean theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles. MA.8.GR.1.2 Apply the Pythagorean theorem to solve mathematical real-world problems involving the distance between two points on a coordinate plane.

4.1 Plotting Integers Using One-Unit Scales for Axes

Lesson Introduction

 Benchmark	MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y -coordinate.		 Learning Target	I can plot with integers using one-unit scales for axes.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.		N/A	
 Essential Questions	- How can you use ordered pairs to locate points in the coordinate plane? - How do coordinate grids organize information?			
 Lesson Narrative	Students investigate the relationship between a point's location in the coordinate plane and the structure of its coordinates. Students then draw conclusions about the signs of the coordinates for ordered pairs in different quadrants. Then students explore how a reflection over the x - or y -axis is performed and its effects on the coordinates of a point. This lesson is accompanied by a Desmos Activity Builder.			
 Component Summary	4.1.1 Warm-Up (~5 min.)	Students predict the number of pieces of candy corn shown in a picture (<i>SE p. 131</i>)	4.1.4 Guided Instruction (10-15 min.)	Students plot ordered pairs and investigate the coordinates of reflections on the coordinate plane (<i>SE p. 135</i>)
	4.1.2 Exploration (10-15 min.)	Desmos Activity Builder connecting ordered pairs to their location on the coordinate plane (<i>SE p. 132</i>)	4.1.5 Guided Practice (5 min.)	Practice identifying the coordinates of reflections of coordinate points (<i>SE p. 136</i>)
	4.1.3 Bring It Together (5-10 min.)	Summary activity on how to read, locate, and write ordered pairs (<i>SE p. 134</i>)	4.1.6 Wrap-Up (~5 min.)	Formative assessment on identifying the coordinates of ordered pairs and their reflections (<i>SE p. 137</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Coordinate plane - Quadrant - Ordered pair		(Optional) Tape for axes (Optional) Highlighters	4.1.1 Warm-Up includes a video to accompany the questions. - Desmos Activity Builder is available to accompany this lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may plot ordered pairs (y, x) instead of (x, y). - Students may have trouble distinguishing reflecting over the x-axis versus the y-axis.	The Desmos Activity provides an opportunity for students to digitally interact with the content and each other. The activity builder could be utilized in a whole group setting, for students to explore in small groups, or a combination of both.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 4.1.2 Exploration positions students to notice the coordinates of points that lie on the x - and y -axes. Consider applying Notice and Wonder routines on each set of images, positioning students to discover the relationship between a point's coordinates and its location in the coordination plane.	MA.K12.MTR.4.1: Discussion and Reflection In the 4.1.2 Exploration, students will engage in mathematical discussion with their peers about the coordinate grid and placement of ordered pairs. Students work with their partner to summarize the characteristics of points in each quadrant in writing. Clarifications within this standard emphasize developing students' abilities to justify their thinking and compare their responses to responses of peers.
 Differentiate Instruction	Prompt students to write out the coordinates of a point they must consider on the coordinate plane. By adding this step to an activity, students have an opportunity to self-check that they have plotted a point correctly. Furthermore, it helps them to solidify connections between a point's location, coordinates, and quadrant. 4.1.2 Exploration implements visual entry points into discussion about the coordinate plane. Consider using the interactive coordinate plane features in the Desmos lesson to provide additional visuals.	
Lesson Notes:		

4.1.1 Warm-Up: Guess the Number

Component Overview: Display the image below for the entire class. After students discuss their estimates, play the video showing the accurate number of candy corn pieces. You can also show the still image of the video for a few moments before playing, for students to estimate.



4.1.1 Warm-Up: Guess the Number

A picture of $\frac{1}{4}$ cup of candy corn is shown.



MA.K12.MTR.4.1: Discussion and Reflection

After students make an estimate independently, place students in groups of three (if possible) to discuss their strategies for estimating. Consider structuring the discussion so each student has a set amount of time to explain their thinking and for their partners to summarize the justification. When sharing whole group, consider asking students to share a justification from a peer that they found interesting.

1. Estimate the number of pieces of candy corn that are in the measuring cup.

Answers will vary. I think the cup holds 12 or 20 pieces of candy corn.

2. Ask two classmates the value they estimated. Record their value and write your summary of their reason. You are the only person who should write in the first two columns of the table. Have the person initial next to your summary stating that your summary is correct.

Answers will vary.

Guess	My Summary of Their Reason	Initials
12	<i>Answers will vary. I tried counting what I saw in the picture.</i>	CL
15	<i>Answers will vary. I multiplied what I think each row holds by each column.</i>	SS

3. How close was your estimate to the actual number?

Answers may vary. My estimate was higher by 4 candy corn because I thought the dish held more space.

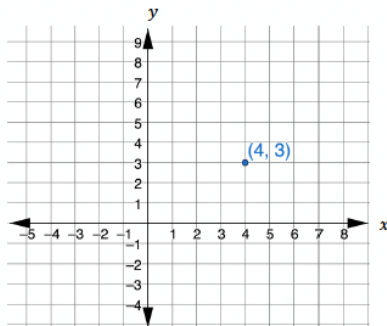
4.1.2 Exploration: The Coordinate Plane

Component Overview: Students will be exploring the relationship between an ordered pair's location and the structure of its coordinates. The first exploration helps students develop the understanding that the x -coordinate represents a point's horizontal position, and the y -coordinate represents a point's vertical position. Then, students investigate the coordinates of ordered pairs located on the axes. Finally, students make a connection between the signs of the coordinates and which quadrants they lie in.



4.1.2 Exploration: The Coordinate Plane

1. This graph shows a point that is plotted on a coordinate plane.



The two numbers are called the coordinates. The two numbers are also known as an ordered pair, (x, y) .

- a. Discuss with your partner which coordinate will change if the point is dragged from left to right.
The x -coordinate will change as the point is dragged from left to right. On a horizontal number line the numbers increase as you move to the right and decrease as you move to the left.
- b. Discuss with your partner which coordinate will change if the point is dragged up or down.
The y -coordinate will change as the point is dragged up and down. On a vertical number line, the numbers increase as you move up and decrease as you move down.



Consider providing sentence stems to help structure partner discussion.

"As the point moves from left to right, the _____ coordinate will _____."

"As the point moves up and down, the _____ coordinate will _____."

4.1.2 Exploration: The Coordinate Plane (continued)

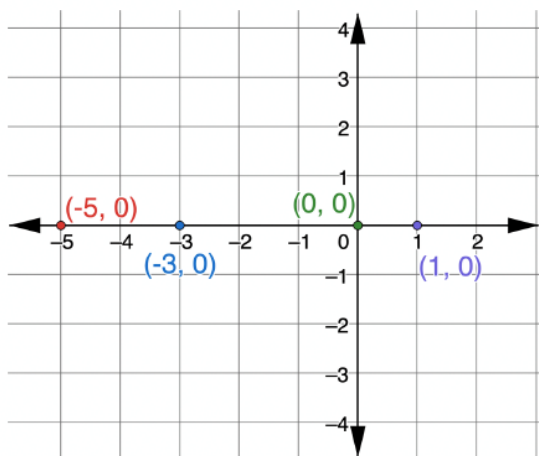
- c. If the point $(4, 3)$ could be dragged around on the coordinate grid, what changes would you expect to see in the ordered pair? Select all that apply.

- ☐ The x -coordinate becomes smaller when the point is moved to the left.
- ☐ The x -coordinate becomes larger when the point is moved to the left.
- ☐ The x -coordinate stays the same when the point is moved to the left.
- ☐ The y -coordinate becomes smaller when the point is moved up.
- ☐ The y -coordinate becomes smaller when the point is moved down.
- ☐ The y -coordinate stays the same when the point is moved up.



Allow students time to discuss incorrect answers with a partner. Also, prompt students to think of edits they could make to some answer choices (like the second and fourth choice) that would make them correct.

2. Four points are plotted on the coordinate grid.



Consider presenting the animation found in the Desmos Activity for question 2. Also consider turning the classroom into a coordinate plane using tape to create the axes. Allow students to walk along the axes and throughout the quadrants to help build understanding.

- a. What do you notice about all four points?
Answers will vary. All of the four points have the same y -coordinate of 0 and are on the x -axis.

4.1.2 Exploration: The Coordinate Plane (continued)

- b. What is the same about the points?

Answers will vary. All of the four points have the same y -coordinate of 0 and are on the x -axis.

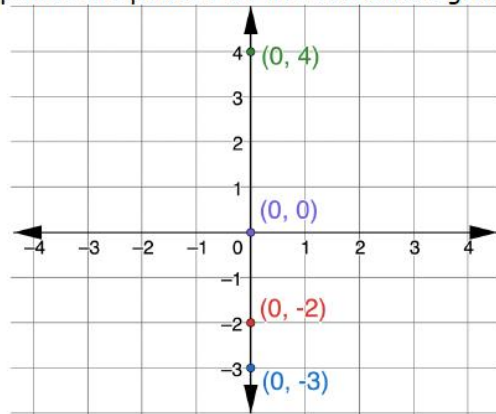
- c. What is different about the points?

All four points have different x -coordinates.

- d. Discuss with your partner what is true about all the points on the x -axis.

All points on the x -axis have the same y -coordinate, and the y -coordinate will be 0.

3. Four points are plotted on the coordinate grid.



- a. What is the same about the points?

Answers will vary. All four points are on the same line, the y -axis, and have the same x -coordinate.

- b. What is different about the points?

Answers will vary. Each point has a different y -coordinate.



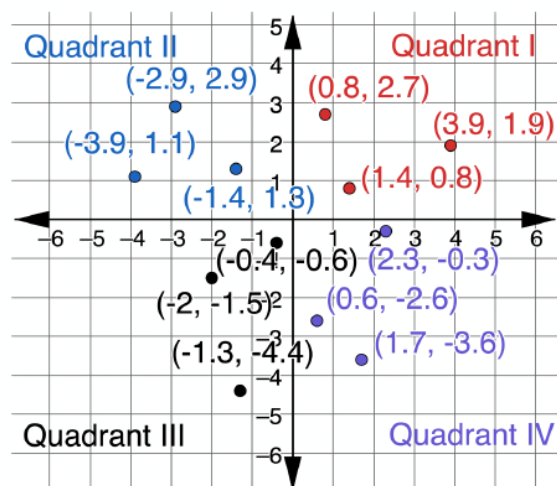
Notice and Wonder

Engage students in a Notice and Wonder routine to position them to draw conclusions about plotting points in the coordinate plane.

- What do you notice?
- What is the same? What is different?
- What does that make you wonder or think about the coordinates of a point either on the x -axis or on the y -axis?

4.1.2 Exploration: The Coordinate Plane (continued)

4. Twelve points are plotted on the coordinate grid. The grid shows the labels Quadrant I, Quadrant II, Quadrant III, and Quadrant IV.



Work with your partner to summarize the characteristics of the points in each quadrant. Write your summary in the table.

Answers will vary.

Quadrant I	<i>The x- and y-coordinates in this quadrant are both positive.</i>
Quadrant II	<i>The x-coordinates in this quadrant are negative. The y-coordinates are positive.</i>
Quadrant III	<i>The x- and y-coordinates in this quadrant are both negative.</i>
Quadrant IV	<i>The x-coordinates in this quadrant are positive. The y-coordinates are negative.</i>



Consider allowing students alternate ways to record their summaries in the table, aside from a written explanation. For example, provide a template like this: (____, ____) so students can record the signs of the coordinates in the appropriate spots.

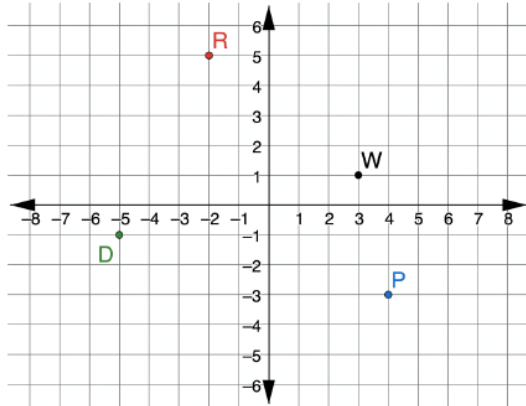
4.1.3 Bring It Together: Plotting Ordered Pairs

Component Overview: Students identify the x - and y -coordinates of ordered pairs and their corresponding quadrants. Students then plot ordered pairs after being given instructions about their coordinates or location.



4.1.3 Bring It Together: Plotting Ordered Pairs

- Points D , P , R , and W are plotted on the coordinate grid.



Select all of the descriptions of each point.

Point D	Point P
☐ The x -coordinate of D is -5 .	☐ The x -coordinate of P is -3 .
☐ The x -coordinate of D is -1 .	☐ The x -coordinate of P is 4 .
☐ The y -coordinate of D is -5 .	☐ The y -coordinate of P is -3 .
☐ The y -coordinate of D is -1 .	☐ The y -coordinate of P is 4 .
☐ Point D is in quadrant I.	☐ Point P is in quadrant I.
☐ Point D is in quadrant II.	☐ Point P is in quadrant II.
☐ Point D is in quadrant III.	☐ Point P is in quadrant III.
☐ Point D is in quadrant IV.	☐ Point P is in quadrant IV.
Point R	Point W
☐ The x -coordinate of R is -2 .	☐ The x -coordinate of W is 1 .
☐ The x -coordinate of R is 5 .	☐ The x -coordinate of W is 3 .
☐ The y -coordinate of R is -2 .	☐ The y -coordinate of W is 1 .
☐ The y -coordinate of R is 5 .	☐ The y -coordinate of W is 3 .
☐ Point R is in quadrant I.	☐ Point W is in quadrant I.
☐ Point R is in quadrant II.	☐ Point W is in quadrant II.
☐ Point R is in quadrant III.	☐ Point W is in quadrant III.
☐ Point R is in quadrant IV.	☐ Point W is in quadrant IV.



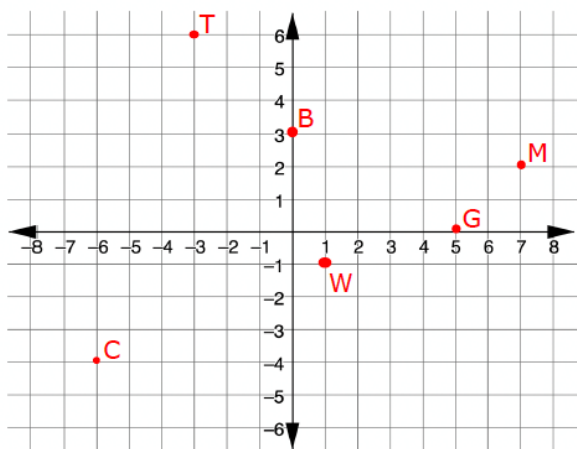
Encourage students to share out explanations for the selections in this activity. Use this as an opportunity to clarify misconceptions about whether the x - or y -coordinate comes first in an ordered pair.



Provide additional auditory entry points for this section to ensure all students have access to the content. It may be beneficial to have students work in pairs to read the descriptions aloud or to pull a small group of struggling readers to work with while the rest of the class works independently.

4.1.3 Bring It Together: Plotting Ordered Pairs (continued)

2. Use the coordinate grid to plot and label points that have the characteristics described in the table.



Answers will vary.

Point B has an x -coordinate of 0. <i>Example shown on graph above.</i>	Point C has a y -coordinate of -4 and is in Quadrant III. <i>Example shown on graph above.</i>
Point G has a y -coordinate of 0. <i>Example shown on graph above.</i>	Point M has a y -coordinate of 2 and is in Quadrant I. <i>Example shown on graph above.</i>
Point T has an x -coordinate of -3 and is in Quadrant II. <i>Example shown on graph above.</i>	Point W has an x -coordinate of 1 and is in Quadrant IV. <i>Example shown on graph above.</i>



If you notice students struggling to plot the points, ask questions to provide further scaffolding:

- Where does a point fall if it has an x -coordinate of zero? Why?
- Where does a point fall if it has a y -coordinate of zero?
- How can you apply what you know about the quadrants to determine where the points should be plotted?



Encourage students to label each quadrant before plotting the points to ensure they can accurately determine where each point should go.

4.1.4 Guided Instruction: Reflections

Component Overview: Students investigate reflections of ordered pairs on the coordinate plane. First, they examine the locations of ordered pairs that share either an x -coordinate or a y -coordinate. Then students reflect ordered pairs over the x - and y -axis.

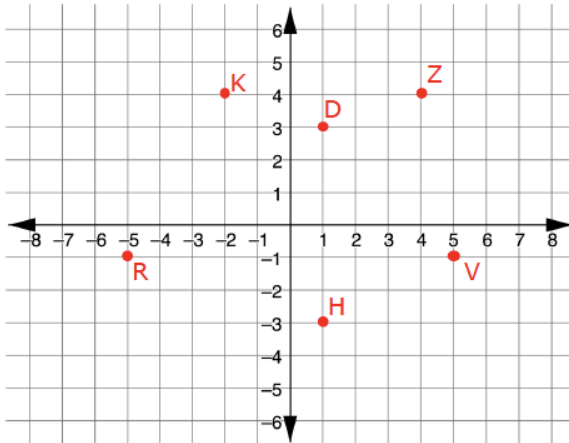


4.1.4 Guided Instruction: Reflections

1. Use the points in the table for the following.

Point D (1, 3)	Point H (1, -3)	Point K (-2, 4)
Point R (-5, -1)	Point V (5, -1)	Point Z (4, 4)

- a. Plot the ordered pairs from the table on the coordinate grid below. Label using the given letter.



- b. Select the pair of points that have the same x -coordinate.

☐ D ☐ H ☐ K ☐ R ☐ V ☐ Z

- c. Select the pair of points that have a y -coordinate of -1.

☐ D ☐ H ☐ K ☐ R ☐ V ☐ Z



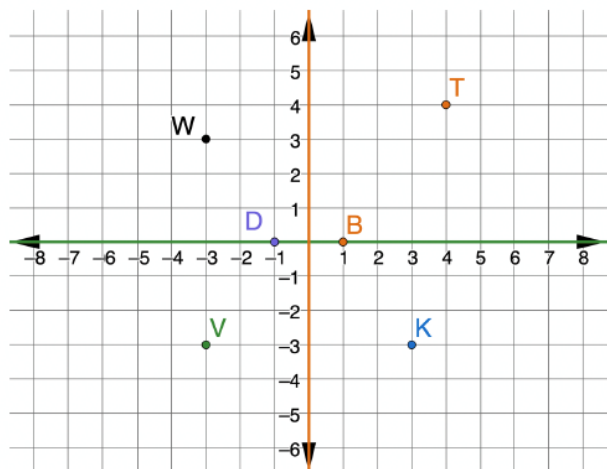
Notice and Wonder

Engage the class in a Notice and Wonder routine to help students draw conclusions about plotting ordered pairs on the coordinate grid.

- What do you notice about points with the same x -coordinate?
- What do you notice about points with the same y -coordinate?
- What do you notice about the distance of these points from the axes?

4.1.4 Guided Instruction: Reflections (continued)

2. A coordinate grid with points D , B , K , T , V , and W is shown. The x -axis is green and the y -axis is orange.



- Write the ordered pair of the reflection of Point V when the graph is folded across the x -axis.
 $(-3, 3)$
- Write the ordered pair of the reflection of Point K when the graph is folded across the y -axis.
 $(-3, -3)$
- Write the ordered pair of the reflection of Point D when the graph is folded across the y -axis.
 $(1, 0)$
- Write the ordered pair of the reflection of Point B when the graph is folded across the x -axis.
 $(1, 0)$



Ask students to discuss strategies to find the reflection of Point T over the y -axis without folding the plane. Once students determine a strategy, prompt them to see if the strategy works for reflections over the x -axis as well.

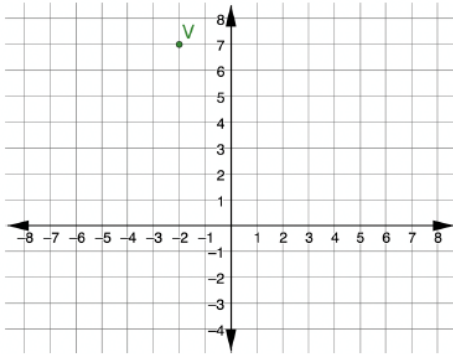
4.1.5 Guided Practice: Reflections

Component Overview: Students practice reflecting ordered pairs over the x - and y -axis.



4.1.5 Guided Practice: Reflections

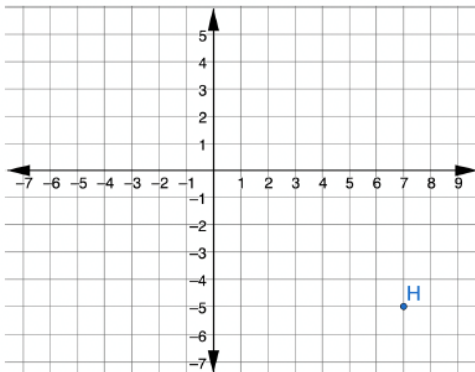
- Point V is plotted on the coordinate grid.



What are the coordinates of the reflection of V across the y -axis?

(2, 7)

- Point H is plotted on the coordinate grid.



What are the coordinates of the reflection of H across the x -axis?

(7, 5)



Use this section as an opportunity to clarify misconceptions before moving on to the 4.1.6 Wrap-Up. Use checks for understanding, like asking where either the points or their reflections are located for the coordinates of the plotted points and the quadrants.



For an additional challenge, ask students to predict what will happen to the coordinates of an ordered pair after reflecting over both axes.

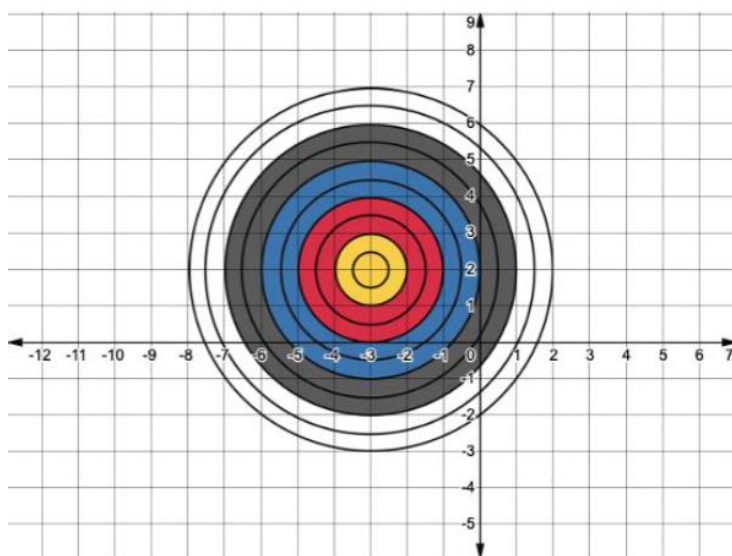
4.1.6 Wrap-Up: Ticket Out the Door

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' abilities to plot an ordered pair. Students will also be reflecting an ordered pair over the x -axis.



4.1.6 Wrap-Up: Ticket Out the Door

A target is shown on the coordinate grid.



Clarify that the bullseye is the very center of a target for students who may be unfamiliar with this vocabulary. In this case, it is the center of the smallest circle in the diagram.

- Write the ordered pair that will hit the bullseye.
 $(-3, 2)$
- A player throws two darts. The first dart hits the ordered pair $(-2, -1)$. The second dart hits the point that is the reflection of $(-2, -1)$ over the x -axis. Complete the table that describes the location of the second dart.

Color	Ordered Pair
Red	$(-2, 1)$

4.2 Points on the Coordinate Plane – Part 1

Lesson Introduction

 Benchmark	MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y -coordinate.		 Learning Targets	- I can plot integers using different scales. - I can plot rational numbers in the form of decimal and fractional ordered pairs. - I can select an appropriate scale given a set of ordered pairs.	
 Benchmark Coherence	Building On		Working Toward		
	MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.		N/A		
 Essential Questions	- How can you determine the scale for ordered pairs that include decimals? - How can you determine the scale for ordered pairs that include fractions? - How can you use the least common multiple to determine the scale for ordered pairs that include fractions?				
 Lesson Narrative	In Lesson 1, students plotted points with integer values on one-unit scales. This lesson builds on what students learned previously to include plotting integer coordinates on a set of axes using different scales. Students will practice choosing a correct scale with decimal units and fractional units and work collaboratively to plot fraction and decimal numbers on the coordinate plane.				
 Component Summary	4.2.1 Warm-Up (~5 min.)	Career profile of an interior designer (<i>SE p. 138</i>)	4.2.4 Collaboration (10 min.)	Plot decimals on a coordinate plane (<i>SE p. 141</i>)	
	4.2.2 Refresher (5 min.)	A reminder of common, but foundational, key terms. (<i>SE p. 139</i>)	4.2.5 Collaboration (10 min.)	Determine best scale to use and plot rational numbers (<i>SE p. 142</i>)	
	4.2.3 Guided Instruction (10-15 min.)	Plot points on a fractional scale (<i>SE p. 139</i>)	4.2.6 Try it (5 min.)	Create a scale to plot decimal points and plot points on a fractional scale independently (<i>SE p. 144</i>)	
	4.2.7 Wrap-Up (~5 min.)	Reflect on understanding of learning targets (<i>SE p. 145</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Coordinate plane - Axes - Coordinate - Origin - Quadrant - Scale - Least Common Multiple Additional Terms: N/A		(Optional) Dry erase boards (Optional) Dry erase markers - Graph Paper		4.2.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with estimating the location of rational ordered pairs. - Students may not understand how to find the least common denominator. - Students may struggle with linking the scale to the least common denominator.	- Help students create appropriate graphs and scales by asking for the largest and smallest x and y values that need to fit in the allotted space. - Students are still grasping rational numbers as well as negative numbers. Consider reviewing what these numbers look like in a visual context to help understanding. - If time is an issue, consider assigning 4.2.6 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 4.2.3 Guided Instruction provides an opportunity for students to engage in a Think-Pair-Share routine to formalize their explanations with a partner before engaging with the written-response questions.	MA.K12.MTR.4.1: Discussion and Reflection This lesson encourages students to work collaboratively with their partners to reason about plotting points on a coordinate plane. Encourage students to analyze their partner's explanations and critique or build upon responses as necessary. Through partner work, students engage in mathematical reasoning and have the opportunity to justify and explain their work.
 Differentiate Instruction	Highlight and make space for the differentiated approaches to sensemaking, which are embedded throughout this lesson. - For verbal processors, or students who learn best when thinking and reasoning with others, the 4.2.4 Collaboration will drive their learning and offer opportunity for making meaning of fluency with scales. - For students who learn best with trial and error or seeing the work in action, consider providing a graphing utility as they work through the 4.2.4 Collaboration.	
Lesson Notes:		

4.2.1 Warm-Up: Interior Designer

Component Overview: In this Warm-Up, students will learn about a career in interior design. After watching a video showing Soledad Alzaga discussing her career, students will consider what career they are interested in and what skills they need for that career.



4.2.1 Warm-Up: Interior Designer

After getting a degree in graphic design, Soledad Alzaga built her career and company as an interior designer. Interior designers make interior spaces functional, safe, and beautiful by determining space requirements and selecting decorative items such as colors, lighting, and materials. Beyond being creative and having good taste, Soledad says to own a successful interior design business, it is necessary to have the ability to work and communicate with others, be organized, have business knowledge, and to have math and measurement skills.

1. What is something that interests you that you would like to spend your career doing?

Answers will vary. I am interested in becoming a Youtuber/Influencer.

2. What skills do you think are needed to be successful in that career?

Answers will vary. I need to have marketing skills and skills for making videos.



Do you know someone who is an interior designer or works in graphic design? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

4.2.2 Refresher: Key Terms With the Coordinate Plane

Component Overview: Students review key vocabulary from Lesson 1 and Grade 5 related to the coordinate plane.

4.2.2 Refresher: Key Terms with the Coordinate Plane

Several key terms related to the coordinate plane were used throughout Lesson 1. Below, you can find definitions for each of these key terms.



The **coordinate plane** is a plane determined by two perpendicular number lines, called axes. The axes intersect at the origin. Each point in the coordinate plane is represented by a pair of coordinates that represent the direction and distance from each axis. The origin is represented by the coordinate pair, $(0, 0)$.



The **axes** of a graph are the horizontal and vertical number lines used in a coordinate plane system.



A **coordinate** is a number used to locate a point on a number line. A coordinate is one of the numbers in an ordered pair that locates a point on a coordinate plane.



The **origin** is the point of intersection of the x - and y -axes in a rectangular coordinate system, where the x -coordinate and y -coordinate are both 0.



A **quadrant** is any of the four regions separated by the axes in a coordinate plane.



Consider vocabulary strategies that align to your classroom and your students' learning needs.

- Frayer Model
- Word wall
- Anchor chart



Prompt students to engage in increasingly precise mathematical discussions using the vocabulary provided. The following might be common, informal examples of words to replace:

- Replace “center” with “origin.”
- Replace “part” with “quadrant.”
- Replace “lines” with “axes.”

4.2.3 Guided Instruction: Plotting Points With a Fractional Scale

Component Overview: In this Guided Instruction, students will practice plotting points that are fractions on a coordinate plane. Students will need to create a fractional scale to accurately plot each point. Students will also draw and create their own graph using fractional points. This activity is also available via Desmos.



4.2.3 Guided Instruction: Plotting Points with a Fractional Scale

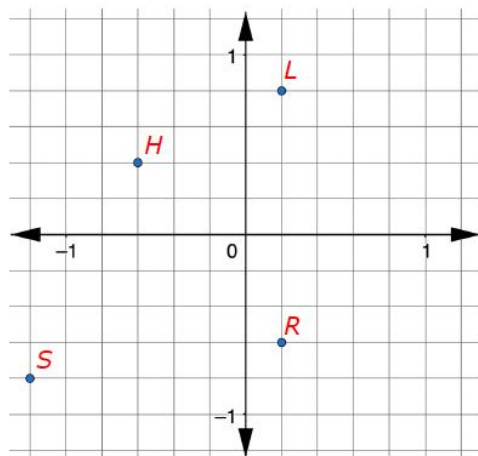
1. Four points are shown.

Point L	Point H	Point S	Point R
$(\frac{1}{5}, \frac{4}{5})$	$(-\frac{3}{5}, \frac{2}{5})$	$(-1\frac{1}{5}, -\frac{4}{5})$	$(\frac{1}{5}, -\frac{3}{5})$

- a. Based on the denominators of each fraction, what scale should be used for the grid?

$\frac{1}{5}$

- b. Find and label the points with the given letters on the coordinate plane.



2. Another set of four points is shown.

Point M	Point L	Point C	Point A
$(\frac{2}{3}, \frac{1}{6})$	$(\frac{1}{2}, -\frac{1}{3})$	$(-\frac{5}{6}, -\frac{1}{2})$	$(-\frac{1}{6}, 1)$

- a. List the denominators of each fraction.

2, 3, 6



Students may struggle to plot the fractional points on the graph. Encourage students to reason about the appropriate scale by asking questions like:

- How many fractional parts should the space between 0 and 1 be broken up into, to accurately plot each point?
- Is there another fractional scale that can be used for this graph?
- Does the scale need to be the same for both the x- and y- axes?

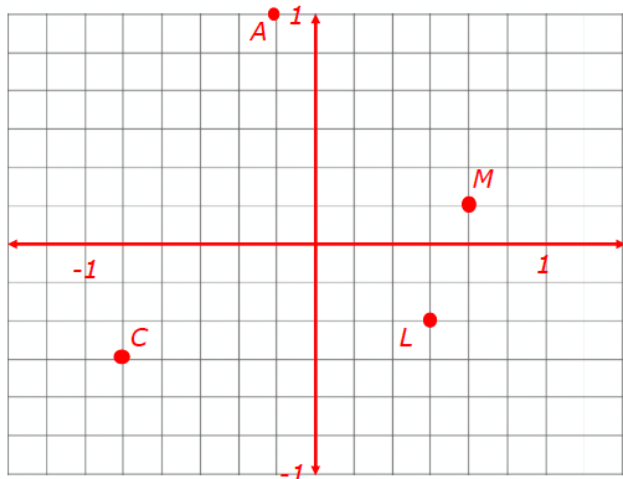
4.2.3 Guided Instruction: Plotting Points With a Fractional Scale (continued)

- b. Zoe says she does not want to label her axes by halves because she will only be able to approximate the location of the thirds. She also says she does not want to label by thirds because there is no easy way to approximate the location of the halves. Christopher says she should label her axes by sixths.

Explain the benefit of using Christopher's plan.

Answers will vary. Christopher's plan will allow him to plot fractions with denominators of 2 and 3 because he will be able to write those as equivalent fractions with a denominator of 6.

- c. Draw and label coordinate axes with the scale you chose and plot the points.



When deciding on a scale for plotting ordered pairs with rational values, consider the least common multiple (LCM) of each of the denominators to determine the proper increments.



Think-Pair-Share

Engage students in a Think-Pair-Share routine as they discuss and complete question 2b. This will give students time to analyze each student's response and determine why Christopher's plan is beneficial. Then students discuss with a partner to encourage mathematical discourse and reasoning.

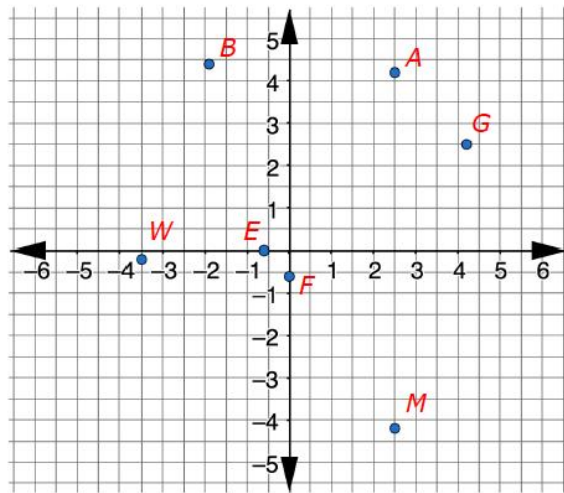
4.2.4 Collaboration: Plotting Points With a Decimal Scale

Component Overview: In this Collaboration, students will work with a partner to plot decimals on a coordinate plane. Students will practice plotting points on a graph that uses whole numbers for the scale and discuss why or why not using this type of graph may work.



4.2.4 Collaboration: Plotting Points with a Decimal Scale

1. A coordinate plane is shown.



- a. Working with your partner, describe the scale that is being used for each axis.

Answers will vary. The scale is 0.5 or $\frac{1}{2}$ because it takes 2 lines to get to each whole number. 1 whole divided by 2 is 0.5 or $\frac{1}{2}$.

- b. Locate each point on the coordinate plane and label it with the appropriate letter.

Point G (4.2, 2.5)	Point B (-1.9, 4.4)	Point F (0, -0.6)	Point W (-3.5, -0.2)
Point E (-0.6, 0)	Point A (2.5, 4.2)	Point M (2.5, -4.2)	



To encourage students to begin to draw conclusions about the scale on the graph, consider asking guiding questions like:

- When graphing Point B, should the x -coordinate be closer to -1 or -2 ?
- What do you notice about where each unit on the scale falls? How many lines apart are they? What does this tell you about the scale?
- What do you know about points that don't fall directly on the lines where the numbers are?

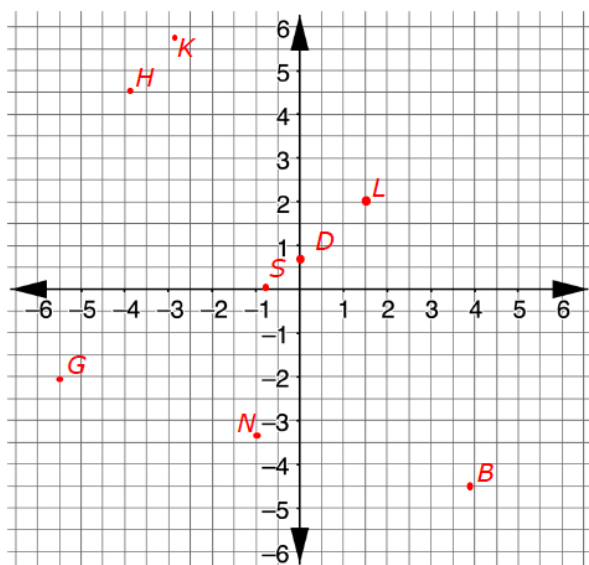
4.2.4 Collaboration: Plotting Points With a Decimal Scale (continued)

2. Working with your partner, plot and label the points using the given letters on the grid.

Point <i>L</i> (1.5, 2)	Point <i>B</i> (3.9, -4.5)	Point <i>G</i> (-5.5, -2.1)	Point <i>D</i> (0, 0.7)
Point <i>K</i> (-2.8, 5.7)	Point <i>S</i> (-0.8, 0)	Point <i>N</i> (-1, -3.4)	Point <i>H</i> (-3.9, 4.5)



Consider allowing students to graph the points using a graphing utility if they are struggling to plot the points on the graph.



- a. Which points were easy to plot?
Answers will vary. Point L, Point D, Point H and Point S
- b. Which points were more difficult to plot?
Answers will vary. Point B, Point G, Point N, Point K



Specifically have students make connections to which points were easy to plot and which were more difficult. Encourage students to reason about what scales might be appropriate for decimal numbers and fractions, making note that points that were not exactly halfway between two numbers might have been more difficult to plot.

4.2.4 Collaboration: Plotting Points With a Decimal Scale (continued)

3. On your own, answer the following question:

When plotting ordered pairs with decimal values, how is it best to determine the scale to use when setting up the axes on a grid?

My Reasoning

Answers will vary. To determine the scale to use when plotting ordered pairs with decimals determine any common factors in the coordinates. For example, if all of the x -coordinates have one decimal place and the tenths place is 0 or 5, then the scale I would choose would be .5 or $\frac{1}{2}$.

Share your reasoning with your partner and ask for feedback on any parts that were unclear or missing. Make any necessary edits to your reasoning.



Consider providing students with sentence starters to help scaffold their thinking on their own and when speaking with their partner.

- “To determine the best scale to use when plotting decimal values, you can...”
- “I know how to choose which scale to use by considering...”

4.2.5 Collaboration: Choosing the Right Scale

Component Overview: In this Collaboration, students work with a partner to determine appropriate scales for graphs based on the data presented. Students choose a graph to best represent a data set and explain why their choice is the best and why the other graphs are not as accurate.



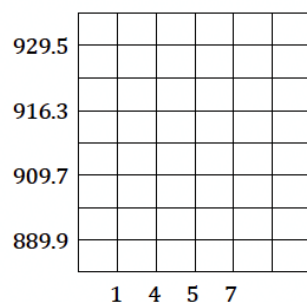
4.2.5 Collaboration: Choosing the Right Scale

When representing data, the scale of the grid is important to effectively and accurately communicate information.

- The table shows the total Florida tourism industry employees, in thousands, for years since 2002.

Years Since 2002	Total Tourism Industry Employees (in thousands)
1	889.9
4	909.7
5	916.3
7	929.5

To plot the data, Andrea drew the coordinate grid shown.



Discuss with your partner why Andrea's coordinate grid would not effectively nor accurately communicate the information about Florida tourism industry employees.

Answers will vary. The scale would not accurately communicate information because it is not a consistent scale. For example, some of the units on the x-axis represent 1 year but some of the units represent 3 years.



MA.K12.MTR.6.1: Reasonableness

Students will discuss why the given graph is not appropriate for the data with a partner. Prompt students with questions like:

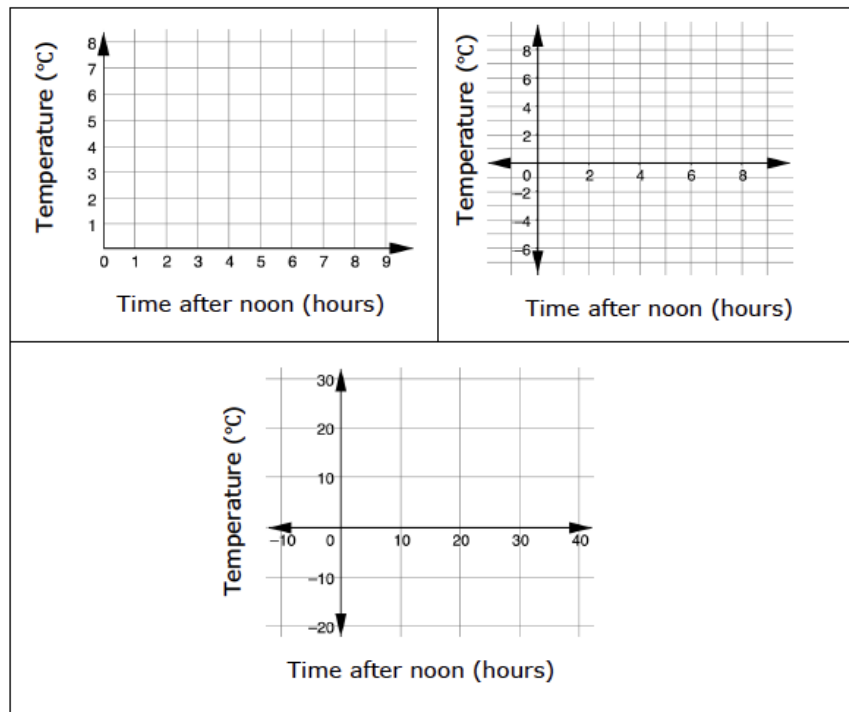
- What is the scale given on the graph?
- How would the graph look when the points are plotted?
- When graphed, would the data accurately show the growth?
- What would be a good scale to use for this set of data?

4.2.5 Collaboration: Choosing the Right Scale (continued)

2. The data shown in the table are average autumn temperatures, in degrees Celsius ($^{\circ}\text{C}$), in Carmathen, Wales.

Time After Noon (hours)	Temperature ($^{\circ}\text{C}$)
0	5
1	4
2	3
3	2
4	1
5	-2
6	-3
7	-4
8	-4

Three coordinate grids are shown, each using different scales.



Think-Pair-Share

Leverage the Think-Pair-Share instructional routine to facilitate meaningful discourse using the prompts in questions 2a and 2b below. Prompt students to individually write their responses to parts a and b before sharing with a partner. Give students the opportunity to refine their responses based on what their partner shared before eliciting responses whole group. Specifically listen for students who change their responses based on what their partner shared and have those students share to continue encouraging growth mindset.

4.2.5 Collaboration: Choosing the Right Scale (continued)

- a. Explain which grid most accurately represents the data.

Answers will vary. The grid in the upper-right corner most accurately represents the data because the x -axis uses a scale that includes all numbers between 0 through 8 while the y -axis includes all the numbers between -4 and 5, which are the ranges of values listed in the table above.

- b. Explain why the other two grids may not be as accurate.

Answers will vary. The grid in the upper left corner is not accurate because it does not include negative values for temperature. The grid at the bottom is not accurate because the scale is too large to accurately plot any of the data and to see any relationships or trends.



As students consider the coordinate grids shown, circulate and ask questions to aid in their analysis like:

- Look at each graph. Are there any points you cannot plot?
- When you graph the data, does it look different on each graph? Does this matter? Why?
- What do you notice about the scales on each of these coordinate grids? Why might that be important when plotting the data points?

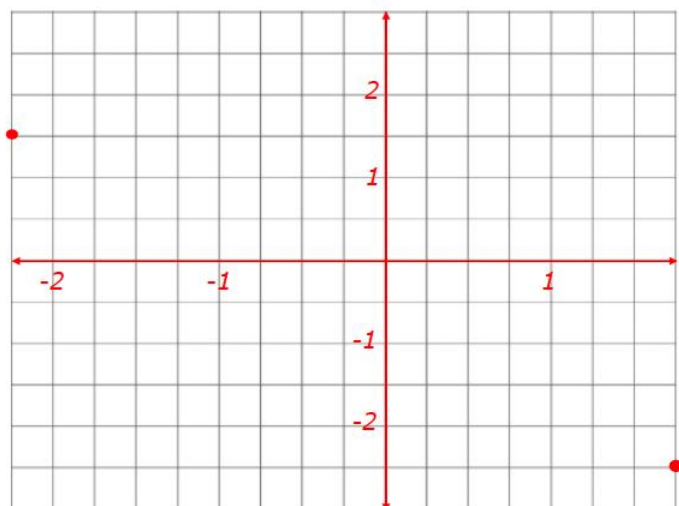
4.2.6 Try It: Plotting Points With Fractional and Decimal Scales

Component Overview: In this Try It, students will be given a blank graph and will need to create a decimal scale and plot decimal points accordingly. Then, students will plot points using a fractional scale. They will need to decide which fractional amount to use for the graphing by finding a common denominator.



4.2.6 Try It: Plotting Points with Fractional and Decimal Scales

1. For the points $(1.75, -2.5)$ and $(-2.25, 1.5)$, draw and label coordinate axes with an appropriate scale, and plot the points.

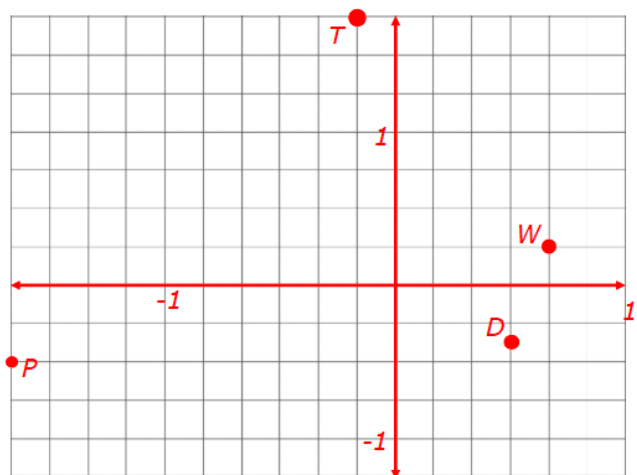


Consider allowing students to plot the points on a dry erase board before writing in their workbooks. This will allow students to use trial and error to arrive at an answer, as they can erase a whiteboard more easily.

4.2.6 Try It: Plotting Points With Fractional and Decimal Scales (continued)

2. Draw and label coordinate axes with an appropriate scale. Then, plot and label the points.

Point <i>W</i>	Point <i>D</i>	Point <i>P</i>	Point <i>T</i>
$(\frac{2}{3}, \frac{1}{4})$	$(\frac{1}{2}, -\frac{1}{3})$	$(-\frac{5}{3}, -\frac{1}{2})$	$(-\frac{1}{6}, \frac{7}{4})$



Students may struggle to determine an appropriate scale because the fractions have different denominators. Further scaffold this section through questioning like:

- What is a common denominator that can be used for all of these coordinates? What's the least common multiple, and how can you find it?
- How can you convert the fractions, so they all have the same denominator?
- Do the *x*- and *y*-axes need to have the same scale? Why or why not?

4.2.7 Wrap-Up: Reflection

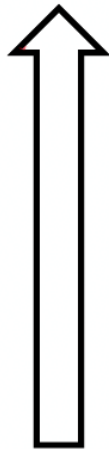
Component Overview: In this Wrap-Up, students will reflect on their understanding of the lesson's learning targets. Consider using this Wrap-Up as a guide for which students will need reteaching or reinforcing of the lessons the next day.



4.2.7 Wrap-Up: Reflection

1. Shade the arrow up to the statement that best describes your current level of understanding.

Answers will vary.



I'm so confident, I could explain plotting ordered pairs made up of fractions and/or decimals to someone else!

I can get to the right answer, but I don't understand it well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked the boxes I think I do not understand.

I tried hard and I listened, but I am finding this challenging. I checked the boxes I think I do not understand.

2. Check the skills that you currently understand.

Answers will vary.

- ☐ Determining the correct scale
- ☐ Identifying the LCM of the denominators for coordinates that include fractions
- ☐ Plotting ordered pairs that contain fractions
- ☐ Plotting ordered pairs that contain decimals



Consider providing additional support for struggling readers by reading the checklist of skills out loud or showing examples of each skill in context.

4.3 Points on the Coordinate Plane – Part 2

Lesson Introduction

 Benchmark	MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y -coordinate.		 Learning Target	I can plot rational numbers using a scale of 1.
 Benchmark Coherence	Building On	Working Toward		
	MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.	N/A		
 Essential Questions	- How can you plot decimal coordinates on a graph with a whole-number scale? - How can you plot fractional coordinates on a graph with a whole-number scale?			
 Lesson Narrative	Students continue plotting coordinates with rational numbers in this second part of a two-part lesson series. Students apply what they have learned in engaging, game-like contexts in addition to real-world examples. Students will need to analyze the data to determine the range of values and the best location of the axes and plot rational numbers with a scale of one.			
 Component Summary	4.3.1 Warm-Up (~5 min.)	Choose the best graph to represent a data set (<i>SE p. 146</i>)	4.3.3 Collaboration (15-20 min.)	Pinpoint coordinates playing a graphing game (<i>SE p. 148</i>)
	4.3.2 Try It (10 min.)	Graph rational numbers (<i>SE p. 147</i>)	4.3.4 Your Turn! (10-15 min.)	Graph rational numbers and compare coordinates (<i>SE p. 149</i>)
	4.3.5 Wrap Up (~5 min.)	Reflect on understanding of past four lessons and discuss next steps (<i>SE p. 150</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Coordinate	(Optional) Computer	The gameboard in 4.3.2 Try It is pre-made and available to print.	

 Teacher Tips	Common Misconceptions - Students may struggle with estimating the location of rational ordered pairs. When graphing negative fractions and decimals on a whole-number scale, students may struggle with which number to graph closer to. - Improper fractions may be incorrectly graphed.	Things to Consider - Students are still grasping negative numbers and may need vertical and horizontal number lines to aid their thinking. - Students may struggle with converting improper fractions to mixed numbers, which may cause a roadblock to graphing. Consider having students convert all mixed numbers before beginning to graph.
	Literacy and Language Routines Notice and Wonder Students participate in a Notice and Wonder routine during this lesson's Warm-Up. This positions students to consider how the scales of graphs impact the way data is presented visually and encourages students to draw conclusions about a data set from a graph.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts 4.3.4 Your Turn! positions students to consider how information in the real world can be represented using graphs and coordinate planes. Students graph data relating to temperature in two cities and then NBA player statistics. Encourage students to draw conclusions about the significance of graphing data in the real world.
	 Differentiate Instruction	
Lesson Notes:		

4.3.1 Warm-Up: Which One Is the Best?

Component Overview: In this Warm-Up, students choose a graph to best represent the data presented and explain why their choice is appropriate.



4.3.1 Warm-Up: Which One Is the Best?

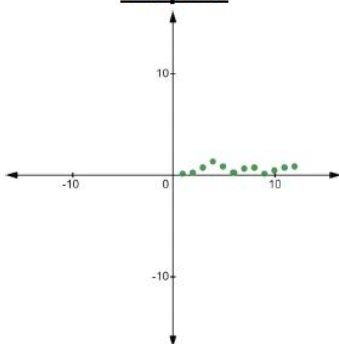
Coral polyps get stressed out and could die if the surrounding water is above 29°C . The table represents the average temperature, to the nearest tenth, above 29°C at the Great Coral Reef over a 12-week period.



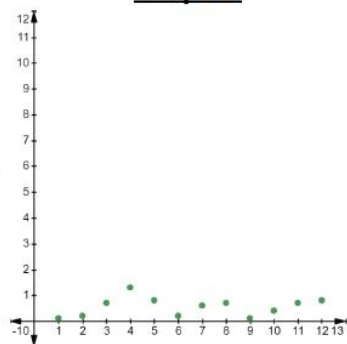
1. Explain which graph best represents the data. (Data taken from NOAA.)

Week	1	2	3	4	5	6	7	8	9	10	11	12
Temp	0.1	0.2	0.7	1.3	0.8	0.2	0.6	0.7	0.1	0.4	0.7	0.8

Graph A



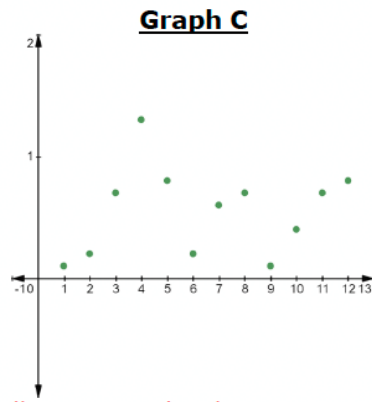
Graph B



This Warm-Up positions students to reason about the appropriateness of scales on coordinate graphs. To help students begin to make connections between the data points and choosing an appropriate scale, ask questions like:

- What do you notice about the temperatures in the table?
- What scale would most clearly show these values? Why?
- How can you use the data points in a set to help determine an appropriate scale?

4.3.1 Warm-Up: Which One Is the Best? (continued)



Answers will vary. Graph C best represents the data because the x-axis accurately represents all the integers used in the data for weeks. The scale for the y-axis depicts the smallest and largest data point to show the variation in the data.

4.3.2 Try It: Game Score

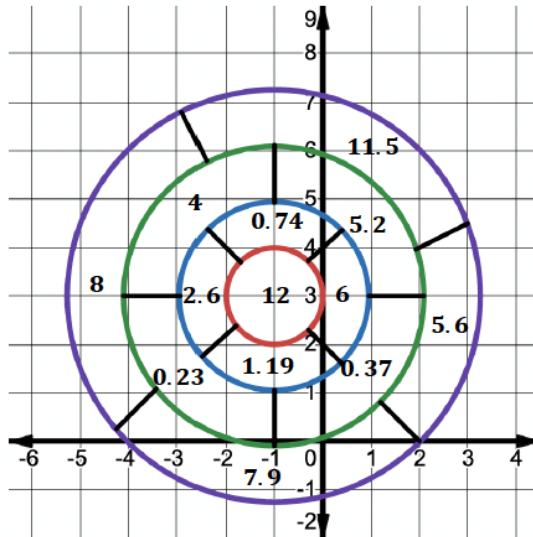
Component Overview: In this Try It, students will plot rational-number coordinates and determine the value of that part of the game board. Students need to plot fractional and decimal coordinates on the given whole-number scale.



4.3.2 Try It: Game Score

A gameboard is shown on the coordinate grid, along with the number of points earned for landing in each area. Each coordinate grid axis is scaled at 1.

- Several points are given in the table. Locate each point on the coordinate grid. Complete the table by writing the score value for landing at each point. Calculate the game score by finding the sum of all the points. Verify your answer with your teacher.



Consider providing additional visual support for students, as the graph may be difficult to read given the number of point values and the scales on the x - and y -axes. It may be helpful to project the game board for the class so students can more clearly see where each point value is in relation to the scales, and/or to color-code each point value with the associated region of the gameboard.



If students are struggling to plot rational numbers on the gameboard, consider further scaffolding through questioning like:

- How can you plot decimal coordinates on a graph that uses a whole-number scale? How can you plot fractions on a graph that uses a whole-number scale?
- Do the coordinates need to be plotted accurately in this activity? Why or why not?

Point	Score Value
$(-3, 4.9)$	4
$(\frac{4}{5}, 1\frac{3}{4})$	0.37
$(0, 2.2)$	6
$(-3.8, 4.9)$	8
$(1.5, 4.4)$	5.2
$(-1\frac{2}{3}, 2)$	1.19
$(\frac{5}{4}, \frac{11}{2})$	11.5

Game Score: 36.26

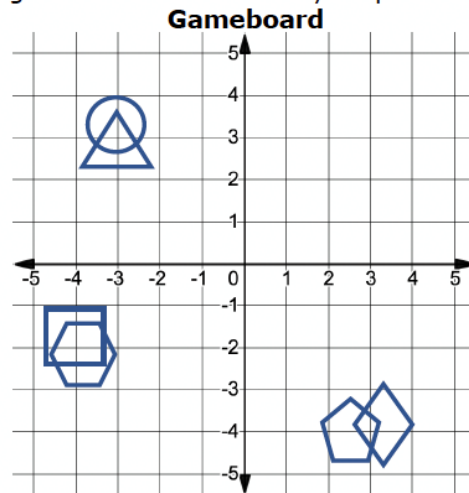
4.3.3 Collaboration: Hit or Miss

Component Overview: In this Collaboration, students will play a Battleship-style game. Each student chooses a shape to plot on their coordinate plane. Each partner guesses a point on the plane to try and determine the shape their partner chose.



4.3.3 Collaboration: Hit or Miss

There are six shapes on the coordinate grid shown. Each partner will select one shape as their game piece. The object of the game is to guess which shape your partner selected before they guess yours. A hit means the point you guessed is inside their shape. A miss means that it is not. Keep your gameboard hidden from your partner during the game.



1. The shape of the region I am picking is a _____.

Answers will vary.

2. Record the coordinates of each guess in the table below. Then, plot the point on the coordinate grid gameboard.

Answers will vary. See example below

Partner 1 Guess	Was it a hit or miss?	Partner 2 Guess	Was it a hit or miss?
(3,5)	Miss	(-3,2)	Miss



Encourage partners to engage in mathematical discussion regarding their strategies.

- Who won? Was their strategy the same as yours?
- Did you have any issues plotting points or coming up with ordered pairs? Why?
- How did your knowledge of graphing points help you in this game?
- How did the scale of the axes impact the guessing game?

4.3.4 Your Turn!

Component Overview: In this Your Turn!, students will plot data on a graph based on real-world contexts. Students plot positive and negative rational numbers on the coordinate plane and use the graphs to draw conclusions about the data set.

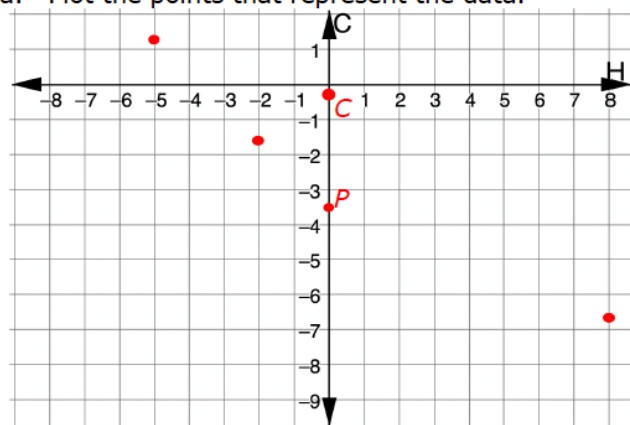


4.3.4 Your Turn!

- The temperature in Princeton, New Jersey was recorded at various times during the day. The times relative to midnight, H , and temperatures in degrees Celsius ($^{\circ}\text{C}$) are shown in the table.

Time (hours before or after midnight, H)	Temperature ($^{\circ}\text{C}$)
-5	$\frac{6}{5}$
-2	$-1\frac{3}{5}$
0	$-\frac{7}{2}$
8	$-6\frac{7}{10}$

- Plot the points that represent the data.



- In the town of New Haven, Connecticut, the temperature at midnight was $-\frac{2}{10}^{\circ}\text{C}$. Plot and label this point on the same grid. Label the point C.
- Label the point that represents the temperature in Princeton at midnight P . Which town was warmer at midnight, Princeton or New Haven?

New Haven is warmer.



Students may graph negative fractions and mixed numbers. Consider posing questions to help students reason about where to plot the points, such as:

- How did you know where to plot the fractions whose denominators were more or less than halfway?
- If the time is in hours before or after midnight, what time is zero?
- Why is 2 the highest number on the y -axis?



If students struggle to answer questions 1c and 1d, consider asking the following to provide scaffolding:

- Which coordinates should be compared to determine the town that was warmer?
- Is a negative mixed number greater or less than a negative fraction?
- As temperatures go further negative, are they getting hotter or colder?

4.3.4 Your Turn! (continued)

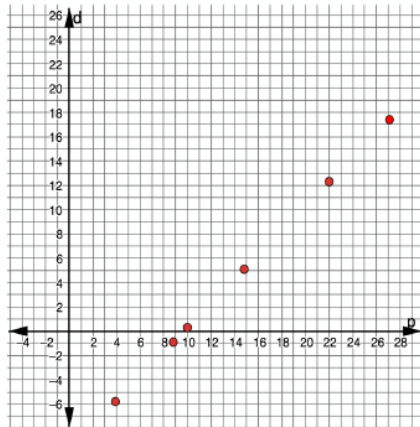
- d. Explain how to determine which town you selected in Part C is warmer at midnight.

Using the y -coordinate of both points, $-\frac{2}{10}$ is further up or greater than $-\frac{7}{2}$ meaning that it is warmer the other number.

2. The average NBA player scores 9.7 points per game. The table below lists six players, along with their average points per game and their difference from the average.

Player	Average Points Per Game (p)	Difference from Average (d)
LeBron James	27.1	17.4
Dwyane Wade	22.0	12.3
Mike Conley	14.8	5.1
Frank Brickowski	10.0	0.3
Robin Lopez	8.8	-0.9
Greg Foster	3.9	-5.8

Graph the points from the table on the coordinate grid.



Students can label each column “ x ” and “ y ” or write out ordered pairs to help remember which axis each number is on.



MA.K12.MTR.7.1: Real-World Contexts

Students are graphing specific NBA players’ average points per game and how much they deviate from the average of the rest of the league. Consider asking:

- What does it mean when a coordinate is close to the x -axis on the graph?
- What does it mean when a player’s coordinates are higher on the graph?
- What does it mean when a player’s coordinates are lower on the graph?



Have students who finish early answer the following:

- What jobs do you think use this type of data?
- How would this graph be helpful to them?
- Are they jobs that you are interested in?

4.3.5 Wrap-Up: Reflection

Component Overview: In this Wrap-Up, students will reflect on their understanding on a scale from one to four to determine what they understand from the past three lessons and how they can improve if their understanding is less than a four. Consider using this reflection to help target who to pull for a small group.



4.3.5 Wrap-Up: Reflection

Think about what you have learned in the past four lessons about plotting points. As you read each statement, consider which best describes your understanding right now.

1. I need help plotting points. I'm not sure where to begin.
2. I can plot points with integers on a coordinate grid with a scale of 1, but I still need help with rational numbers and different scales.
3. I can plot points with decimals on a coordinate grid with a tenths scale, but I still need help with points that are fractions.
4. I can plot points with rational numbers that are decimals or fractions using any scale.

1. Explain which number from the list best describes you.

Answers will vary.

2. Write a plan for what you can do to either move to number 4 or to help a classmate move to number 4.

Answers will vary. I will practice labeling and plotting ordered pairs on a graph to increase my understanding and accuracy.



Students may need to put either a checkmark or an "X" by each statement they agree or disagree with to help frame their thinking. When answering question 2, consider giving students a sticky note to write their answer on and place on their desk to help remind themselves how to achieve their goal.

4.4 Distance on the Coordinate Plane

Lesson Introduction

 Benchmark	MA.6.GR.1.2 Find the distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.		 Learning Target	I can find the distance between two ordered pairs.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.8.GR.1.2 Apply the Pythagorean theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.	
 Essential Questions	- How can you find the distance between two ordered pairs whose y-coordinates are the same? - How can you find the distance between two ordered pairs whose x-coordinates are the same? - How is absolute value related to finding the distance between two ordered pairs?			
 Lesson Narrative	In Lesson 4, students will begin developing their understanding of finding distance on the coordinate plane. Students will build upon knowledge of graphing opposites on both vertical and horizontal number lines. You will lead students toward understanding the relationship between finding the distance between values on the coordinate plane to operations with absolute value.			
 Component Summary	4.4.1 Warm-Up (~5 min.)	Plot rational numbers on a vertical number line (<i>SE p. 151</i>)	4.4.4 Guided Practice (10 min.)	Find distance between two points using addition and subtraction of coordinates (<i>SE p. 155</i>)
	4.4.2 Collaboration (5 min.)	Collaborate with a partner on how to find distance between two points (<i>SE p. 152</i>)	4.4.5 Your Turn! (10-15 min.)	Find the distance between two points (<i>SE p. 156</i>)
	4.4.3 Guided Instruction (15-20 min.)	Count on a grid to find distance between two points (<i>SE p. 153</i>)	4.4.6 Wrap-Up (~5 min.)	Create a summary of how to find distance between two points (<i>SE p. 158</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Absolute value <u>Additional Terms:</u> - Quadrant		(Optional) Graph paper	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may confuse when to add the absolute values and when to subtract the absolute values when finding the distance between two points. For example, when finding the difference between two points, if one is negative, students may believe that negative represents the difference, then add the two values.	- Students have not been exposed to operations with integers prior to this unit. Therefore, students cannot use those operations to find the distance between points with negative coordinates. A foundation for operations with integers can be laid by verbally expressing the subtraction while writing the expression. If students are naturally understanding the operations, then the teacher can further explain. But student understanding of the operations is not necessary for their success in this lesson. - Consider bringing the students back to the visual of the coordinate plane when determining whether to add or subtract to find the distance.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Consider implementing a Think-Pair-Share for 4.4.2 Collaboration. This collaboration positions students to think about how they would find distance between two coordinates and share their thoughts with a partner. Consider supplying a sentence starter such as “I think ____ is correct because...”.	MA.K12.MTR.6.1: Reasonableness When finding distance between two points, students should use reasoning and sensemaking to determine whether their answer is correct or not. Consider encouraging students to check every answer for reasonableness in order to verify their answers. Ask questions like “Does this answer make sense in this context?”.
 Differentiate Instruction	Guess My Point Consider providing students with an alternative way to continue applying their knowledge of the coordinate plane and finding distance with a game of Guess My Point. Similar to <i>Guess Who?</i> , students in pairs must ask one yes or no question at a time to determine the secret location of their partner’s ordered pair. Once both students have discovered the secret location, they must work together to find the distance between their locations. This can be done by mapping your classroom as well, with students picking secret objects around the room that have been designated an ordered pair as a location. Read over 4.4.5 Your Turn! to see an example.	
Lesson Notes:		

4.4.1 Warm-Up: A Point on the Number Line

Component Overview: Students plot rational numbers on a vertical number line.



4.4.1 Warm-Up: A Point on the Number Line

1. Circle all numbers that could be the location of point F on the number line.

3.5

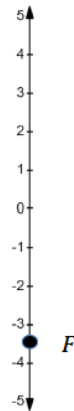
$4\frac{1}{2}$

$-\frac{3}{5}$

-3.5

$-\frac{7}{2}$

$-4\frac{1}{2}$



Connect the vertical number line to the coordinate plane that students have been working with throughout this unit and will continue to work with in this lesson.

- How is the vertical number line similar to the coordinate plane?
- How can you use what you know about number lines to plot points on the coordinate plane?

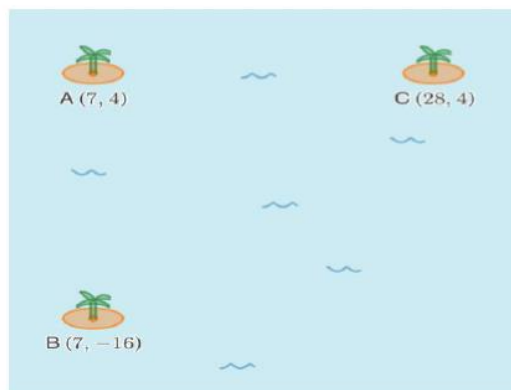
4.4.2 Collaboration: Who Is Correct?

Component Overview: Students will collaborate with a partner to determine the distance between three islands. They are given coordinates of each island and must use those and their mathematical reasoning to determine which student is correct about the islands' distances from one another.



4.4.2 Collaboration: Who Is Correct?

The locations of three islands in the ocean are given using ordered pairs.



Three students described the distance between each island.

Braxton says the distance from point A to point C is **less than** the distance from point A to point B.

Justine disagrees with him and says the distance from point A to point C is **greater than** the distance from point A to point B.

Rainey says the distance from point A to point C is **equal to** the distance from point A to point B.

1. Discuss with your partner who you think is correct. Write a summary of your discussion.

Answers will vary. Justine is correct because Island A is only 20 units away from Island B, while it is 21 units away from Island C.



Think-Pair-Share

Consider implementing a Think-Pair-Share for 4.4.2 Collaboration. This Collaboration positions students to think about how they would find the distance between two coordinates and share their thoughts with a partner.

"I think ____ is correct because..."

Spend time probing students on their reasoning for which student they think is correct. Highlight student answers that use the coordinates and add or subtract to find distance.

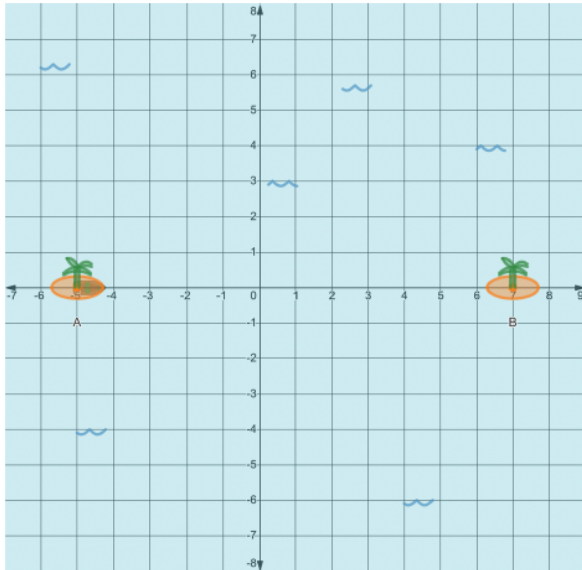
4.4.3 Guided Instruction: Distance on the Coordinate Plane

Component Overview: Students use counting grid lines to find the distance between two islands or two points plotted on the coordinate plane. Students must also be able to identify the coordinates of the points and use those coordinates to discover how to find distances between points on the coordinate plane.



4.4.3 Guided Instruction: Distance on the Coordinate Plane

- The coordinates of two small islands are labeled points A and B on the graph.



- Counting the grid lines, how far apart are the islands?
12 units
- What are the coordinates of points A and B ?
 $A: (-5, 0)$
 $B: (7, 0)$
- Explain why the y -coordinates of the two points are the same.
Answers will vary. The y -coordinates are the same because the islands are horizontal to one another, and both islands are on the x -axis.



Notice and Wonder

Engage students in a Notice and Wonder routine to review the vocabulary and components of the coordinate grid.

- What do you notice?
- What do you wonder about the two islands?
- What axis are the islands on? Label it. Label the other axis.
- Are the islands the same distance from the origin?



Some students may confuse how to write the coordinates correctly. If students write $(0, -5)$, review where the x - and y -coordinates go in an ordered pair (x, y) .

4.4.3 Guided Instruction: Distance on the Coordinate Plane (continued)

- d. Describe the location of the two points relative to the y -axis.

Island A is 5 spaces to the left of the y -axis

Island B is 7 spaces to the right of the y -axis



The distance a number is from zero (0) on a number line is called the **absolute value**.

- e. What is the absolute value of the x -coordinate of each point?

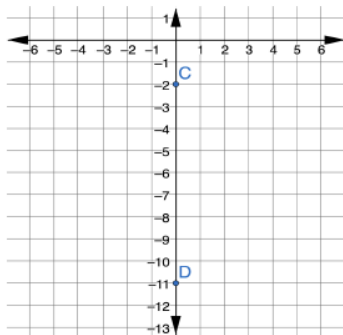
The absolute value of the x -coordinate of Point A is 5.

The absolute value of the x -coordinate of Point B is 7.

- f. Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.

$$|-5| + |7| = 12$$

2. Below is another coordinate plane with two different islands located at points C and D.



- a. How is the location of these two islands different from the first two you worked with?

These islands are both on the y -axis and the other two were both on the x -axis.

- b. Counting the grid lines, how far apart are the islands?

9 units

- c. What are the coordinates of points C and D?

C: (0, -2)

D: (0, -11)



*Solidify student understanding through questions like:
Do you think this will always be true for two points on a horizontal line? Why or why not?*



Students may need a reminder on how to write a mathematical sentence using absolute value.

Consider prompting by writing the following on the board:

$$| \quad | + | \quad | = \quad$$

4.4.3 Guided Instruction: Distance on the Coordinate Plane (continued)

- d. Explain why the x -coordinates of the two points are the same.

Answers will vary. The x -coordinates are the same because the two points are on the same vertical line. The two points are both on the y -axis.

- e. Describe the location of the two points relative to the x -axis.

Point C is 2 spaces below the x -axis

Point D is 11 spaces below the x -axis

- f. What is the absolute value of the y -coordinate of each point?

The absolute value of Point C is 2.

The absolute value of Point D is 11.

- g. Write a mathematical sentence using absolute value that can be used to determine how far apart the two islands are.

$$|-11| - |-2| = 9$$

3. Complete the statements.

When finding the horizontal distance between two points that lie on the same line, if the points are on

opposite sides of the y -axis, ☐ add ☐ subtract the absolute value of the unlike coordinates. If the points are on the

same side of the y -axis, then ☐ add ☐ subtract the absolute values of the unlike coordinates.

When finding the vertical distance between two points that lie on the same line, if the points are on the

opposite sides of the x -axis, ☐ add ☐ subtract the absolute value of the unlike coordinates. If the points are on the

same side of the x -axis, then ☐ add ☐ subtract the absolute value of the unlike coordinates.



Consider creating a 3D number line on the floor and having students walk the number line to count the distance between two points. Ask the class to count the distance out loud. Have students try a few examples where they are on the same side of the number line and a few where they are on different sides of the number line.



Solidify student understanding through intentional questions like:

Do you think this will always be true for two points on a vertical line? Why or why not?



Notice and Wonder

Have students illustrate/draw/sketch as they read the paragraph. Once they have sketched what they read, prompt students to notice the similarities and differences based on location in quadrants.

- What do you notice?
- What is the same? What is different?
- What does that make you wonder or think about when finding distance on the coordinate plane?

Monitor for student responses that include:

- "A pattern I noticed among the coordinates of the points in the same quadrant is..."
- Adding or subtracting the coordinates to find distance.



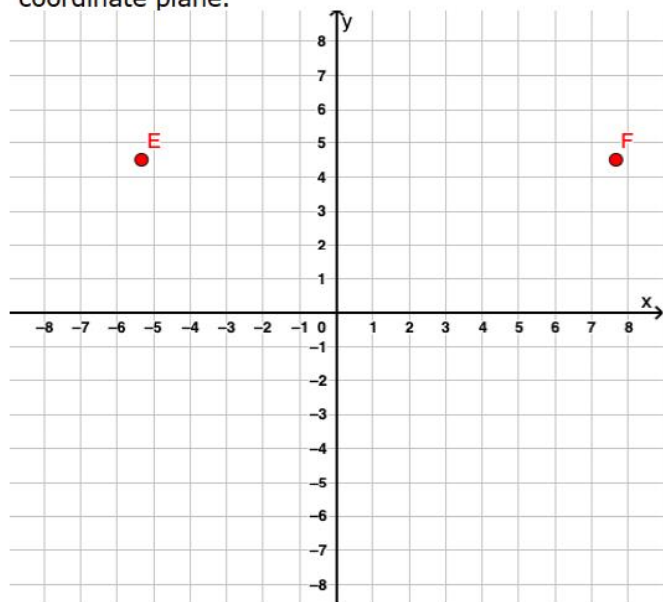
4.4.4 Guided Practice: Distances Between Rational Numbers

Component Overview: Students practice finding distance between two points on the coordinate plane using addition or subtraction.



4.4.4 Guided Practice: Distances Between Rational Numbers

1. Graph the points $E(-5\frac{1}{3}, 4\frac{1}{2})$ and $F(7\frac{2}{3}, 4\frac{1}{2})$ on the coordinate plane.



- a. Describe the location of these points relative to the origin and the quadrants they lie in.

Answers will vary. Point E is $5\frac{1}{3}$ units left and $4\frac{1}{2}$ units up from the origin in quadrant II.

Point F is $7\frac{2}{3}$ units right and $4\frac{1}{2}$ units up from the origin in quadrant I.

- b. What is the distance between point E and point F?

$$\left| -5\frac{1}{3} \right| + \left| 7\frac{2}{3} \right| = 12\frac{3}{3} \text{ or } 13$$



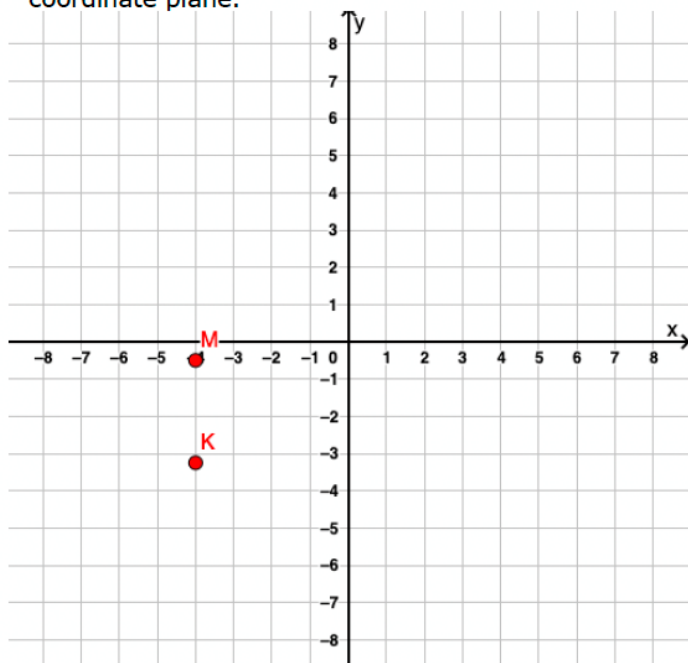
Florida's assessment limits for MA.6.GR.1.2 and MA.6.GR.1.3 state that all ordered pairs should be integers. The rational ordered pairs in 4.4.4 and 4.4.5 help to support the area of emphasis by requiring students to perform all four operations with integers, positive decimals and positive fractions with procedural fluency, while also continuing student understanding of finding distance. By stretching the student's experience beyond integers, the student will not necessarily be able to rely on counting to determine the distance, thus solidifying their understanding of absolute value as the distance from zero.



Check for understanding of plotting fractions on the coordinate plane. Misconceptions might include students plotting $-5\frac{1}{3}$ at $\frac{1}{3}$ units to the right of -5 instead of to its left.

4.4.4 Guided Practice: Distances Between Rational Numbers (continued)

2. Graph the points $K(-4, -3.25)$ and $M(-4, -0.5)$ on the coordinate plane.



Consider giving students only the coordinates and having them find the distance without the visual of the coordinate plane.

- a. Describe the location of these points relative to the origin and the quadrants they lie in.

Answers will vary.

Point K is 4 units left and 3.25 units down from the origin in quadrant 3.

Point M is 4 units left and 0.5 units down from the origin in quadrant 3.

- b. What is the distance between point K and point M ?

$$|-3.25| - |-0.5| = 2.75$$

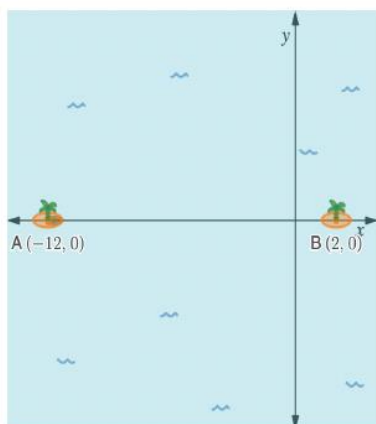
4.4.5 Your Turn!

Component Overview: Students practice finding distance between two points on the coordinate plane using addition and subtraction.



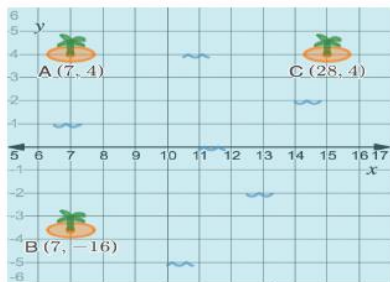
4.4.5 Your Turn!

- What is the distance from point A to point B?



Point A is 14 units away from point B. $|-12| + |2| = 14$

- Explain which island is closest to island A. Show your work.



Answers will vary.

Distance between point A and B: $|-16| + |4| = 20$

Distance between point A and C: $|28| - |7| = 21$

Island B is closest to island A because they are only 20 spaces apart, while island C is 21 spaces from island A.



Notice that in question 1 students don't have the grid lines to count the spaces between the two islands. This will prove difficult for some students who might have been using this method. Prompt those students to determine how they might find the distance without counting. Once they have answered, encourage them to then use the counting method to verify their answer.

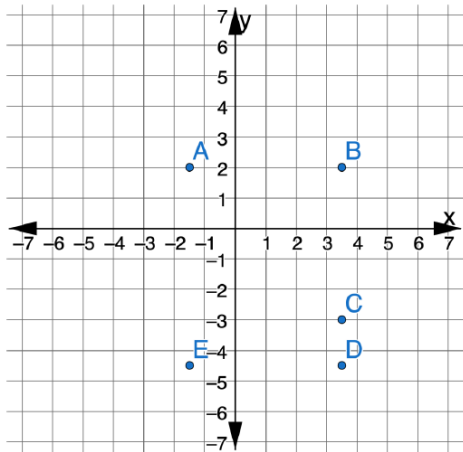


Check for students using mathematical reasoning to explain their answers. Consider providing sentence frames for students to explain their reasoning:

- Island ____ is closest to Island A because ____.
- I used adding/subtracting to show that Island ____ is closest to Island ____ by ____ units.

4.4.5 Your Turn! (continued)

3. Five points are shown. Points A and E lie on the same vertical line. Points B , C , and D lie on a different vertical line. Points E and D lie on the same horizontal line.



- a. Complete the table by filling in each missing coordinate.

A	$(-1.5, \text{ })$
B	$(\text{ } , 2)$
C	$(3.5, \text{ })$
D	$(3.5, \text{ })$
E	$(\text{ } , -4.5)$

- b. The distance between points B and C is 5 units.
- c. The distance between points D and C is 1.5 units.
- d. The distance between points D and E is 5 units.



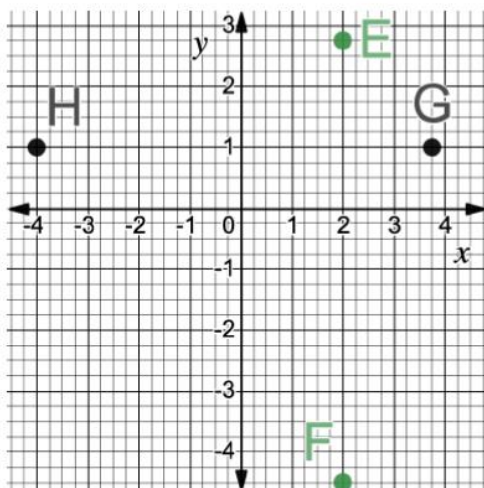
Guess My Point

For students who finish early or for an alternative method for engaging students in the practice of plotting and finding distance, consider having students play the Guess My Point activity.

- Each partner picks a secret location for their point. (Choose an ordered pair.)
- Students then take turns asking one question at a time to determine the location of their partner's point (similar to Guess Who?). Encourage students to ask questions that incorporate vocabulary from the lesson. For example, "Is your point in Quadrant 1?" or "Does your location have a positive x -coordinate?"
- Once both students have discovered the location of their partner's point, they must collaboratively find the distance between those two points.

4.4.5 Your Turn! (continued)

4. Use the coordinate plane below to determine distances between the following points.



Notice the answer here is in fourths. Consider prompting students to identify the scale first when reading the graph. Students should notice that each grid line is an increase of $\frac{1}{4}$.

- The distance between points E and F is 7.25 units.
- The distance between points G and H is 7.75 units.
- Which points are the same distance from $(2, 1)$?
Points E and G are the same distance from the point $(2, 1)$.
Point E is 1.75 units from the point $(2, 1)$.
Point G is also 1.75 units from the point $(2, 1)$.

4.4.6 Wrap-Up: Cool Down

Component Overview: *Students write a summary for a friend about how to find the distance between two points on the coordinate plane.*



4.4.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them learn how to find the distance between two points on a coordinate plane.

Answers will vary.



For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson.

- *Recording a message explaining the concept or examples verbally*
- *Drawing and labeling, providing an example*
- *Constructing a graphic organizer*

4.5 Perimeter on the Coordinate Plane

Lesson Introduction

 Benchmarks	MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane. MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.		 Learning Targets	- I can find the distances between ordered pairs. - I can find the dimensions of a rectangle on a coordinate plane. - I can find the perimeter of a rectangle on a coordinate plane.
 Benchmark Coherence	Building On MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas		Working Toward MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane, involving perimeter or area of polygons.	
 Essential Questions	- How can you use what you know about distance on a coordinate plane to find the dimensions of a rectangle? - How can you use what you know about distance on a coordinate plane to find the perimeter of a rectangle on the coordinate plane?			
 Lesson Narrative	Students apply their knowledge of finding distance on the coordinate plane to help them find the dimensions of rectangles, as well as determining the perimeter of a rectangle. Students should be led to see how using the coordinate plane and what we know about the properties of a rectangle give them more than one way to determine distance and dimensions.			
 Component Summary	4.5.1 Warm-Up (~5 min.)	Complete self-assessment on understanding of topics applied in this lesson (<i>SE p. 159</i>)	4.5.4 Collaboration (15 min.)	Collaborate on an error analysis and work together to fix the mistakes (<i>SE p. 162</i>)
	4.5.2 Guided Instruction (10-15 min.)	Find distance between points and identify dimensions of a rectangle on the coordinate plane (<i>SE p. 160</i>)	4.5.5 Your Turn! (5-10 min.)	Find dimensions and solve for perimeter of a rectangle on the coordinate plane (<i>SE p. 164</i>)
	4.5.3 Try It (5-10 min.)	Solve for perimeter of a rectangle on the coordinate plane (<i>SE p. 162</i>)	4.5.6 Wrap-Up (~5-10 min.)	Find dimensions and solve for perimeter of a rectangle on the coordinate plane (<i>SE p. 165</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Perimeter - Absolute value - Coordinate plane - Origin		(Optional) Graph paper	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may always want to add or subtract the coordinates to find the distance without thinking about the location on the coordinate plane, the relation to the axes, and the relation of the points to each other. - Students may believe the orientation of a rectangle is always the same. For example, students may believe length is always the “sides” of the rectangle. - Students may struggle with the concept of absolute value.	Length is not always the longest side of the rectangle. Unless the length is specified or named, students can choose either side as length and width and still be mathematically correct. 4.5.2 Guided Instruction, 4.5.4 Collaboration, 4.5.3 Try It, and 4.5.6 Wrap-Up all specify which side is the length, but 4.5.5. Your Turn! leaves the length open-ended for students to decide. Consider prompting this discussion during 4.5.2 Guided Instruction to prepare students for the flexibility in their 4.5.5 Your Turn! and 4.5.6 Wrap-Up questions.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 4.5.4 Collaboration positions students to engage in a Think-Pair-Share. Have students think independently first about Vera and Conor’s work and some errors they might notice before having students discuss with their partner.	MA.K12.MTR.4.1: Discussion and Reflection Students engage in an error analysis where they have to determine which student’s work is correct, if any, and also correct that work using mathematical thinking.
 Differentiate Instruction	Consider moments in which some of the coordinates provided (i.e., a mixed number) could be substituted with a friendlier number to allow students to focus on the process of finding distance and perimeter, rather than the ability to add and subtract with fractions. This will provide additional entry points for students to engage with the content. During 4.5.3 Try It, consider providing students with graph paper as an option to draw and label the fence as an additional visual entry point. During 4.5.4 Collaboration, consider allowing students to explain verbally instead of writing.	
Lesson Notes:		

4.5.1 Warm-Up: Self-Assessment

Component Overview: Students use self-assessment to show understanding of previously learned topics that will be applied in this lesson.



4.5.1 Warm-Up: Self-Assessment

1. Complete each statement to describe your understanding.

- ☐ I know
☐ I need support with
- how to plot rational number

ordered pairs in all four quadrants.

Answers will vary.

- ☐ I know
☐ I need support with
- how to find the perimeter and area

of rectangles.

Answers will vary.

- ☐ I know
☐ I need support with
- how to find the vertical or

horizontal distance between two ordered pairs that are

located in the same or different quadrants.

Answers will vary.



Consider positioning students as experts and create partners based on strengths and gaps of students. For example, identify four students who chose “I know” regarding perimeter and area and three students who chose “I know” for finding distance, and create seven small groups with one of those seven students as the captain of each group. All other students can then opt-in to one of the small groups based on their needs. Alternatively, students can pair up with someone around the room and hold their own small groups using Lesson 4 resources.

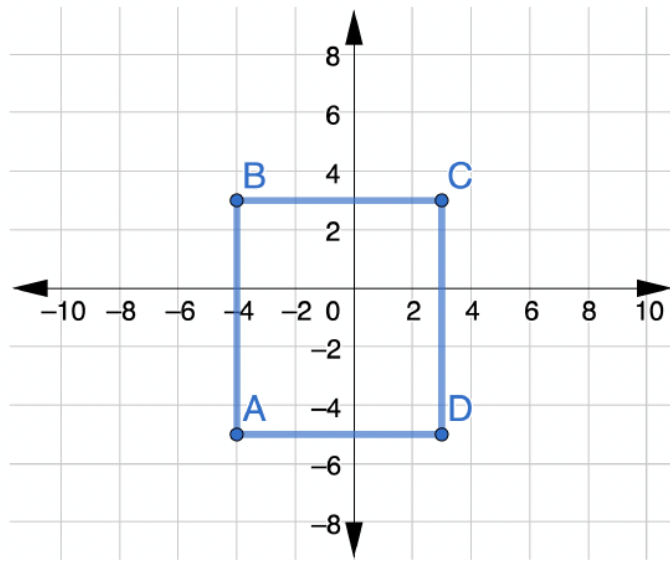
4.5.2 Guided Instruction: Dimensions and Perimeters of Rectangles

Component Overview: Students apply what they know about finding distance between points to find the dimensions of a rectangle on the coordinate plane. Using those dimensions, students apply what they know about perimeter of a rectangle to find perimeter on the coordinate plane.



4.5.2 Guided Instruction: Dimensions and Perimeters of Rectangles

1. A rectangle has vertices at the points $A(-4, -5)$, $B(-4, 3)$, $C(3, 3)$, and $D(3, -5)$ as shown.



- a. The distance between points A and D is the length of the rectangle. The length of the rectangle is 7 units.
- b. What other two points on the rectangle are the same distance apart as points A and D ?
Points B and C are also the same distance apart as points A and D . They are both 7 units apart.

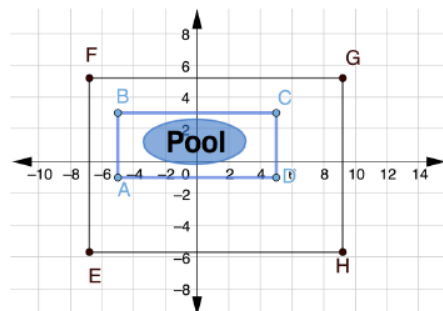


Position students to verbalize how they can find the length and width of the rectangle. Encourage students to recognize that counting the spaces may not necessarily be helpful in this instance because some of the coordinates are not integers. Ensure students understand there is more than one way to find the distance between points on the coordinate plane.

This lesson is a foundation for students understanding how to find slope as the change in the vertical distance over the change in the horizontal distance.

4.5.2 Guided Instruction: Dimensions and Perimeters of Rectangles (continued)

- c. Between which two points is the line segment that represents the width of the rectangle?
Answers will vary. The distance between points A and B represent the width. The distance between points D and C represent the width.
- d. Find the width of the rectangle.
8 units
2. Sonja's family installed a new pool. Sonja's dad wants to put up a fence, shown as a blue rectangle, around the pool for safety. The scale of the grid is represented in meters (m). He needs to know how much fencing he will need.



The rectangle that represents the fence has vertices at $A(-5, -1)$, $B(-5, 3)$, $C(5, 3)$, and $D(5, -1)$.

- a. What is the length of the line segment that connects points A and D?
10 meters
- b. What is the length of the line segment that connects points A and B?
4 meters
- c. Explain how to calculate the perimeter of the fence.
Answers will vary. One way to find the perimeter is two times the length from A to D (width of the pool) plus two times the length from A to B (length of the pool).
- d. What is the perimeter of the fence in meters?
28 meters



Always, Sometimes, or Never

Is the length of a rectangle always, sometimes, or never the longest side?

Length is not always the longest side. Unless the question specifies which side is the length, as seen in this question, students can choose either side as length or width and mathematically be correct, as the perimeter does not change.

4.5.5. Your Turn! questions do not specify which side is the length.



If students are struggling to find the length and width of the rectangles, ask probing questions like:

- How do we find distance between two points?
- Are the points on the same side of the axis or different sides? Does this matter? Why or why not?
- *Are the points on the same side of the axis or on opposite sides? How far are the points from the axis? How can we use this to help us find out how far the points are from each other?*



Expressive Language: Eliminate Barriers

Consider providing sentence frames for students to explain their reasoning (i.e., I am able to find the perimeter of the rectangle by ____.)

4.5.3 Try It: Perimeter on the Coordinate Plane

Component Overview: Students solve for perimeter of a rectangle on the coordinate plane.



4.5.3 Try It: Perimeter on the Coordinate Plane

1. Sonja's father plans to put another fence around the yard. The black rectangle on the coordinate grid in the previous component represents the second fence. The scale of the grid is represented in meters (m). The black rectangle has vertices at $E (-6.8, -5.7)$, $F (-6.8, 5.2)$, $G (9.2, 5.2)$, and $H (9.2, -5.7)$.
 - a. What is the length of the line segment that connects point E to point H ?
16 meters
 $|-6.8| + |9.2| = 16 \text{ meters}$
 - b. What is the length of line segment that connects point E to point F ?
10.9 meters
 $|-5.7| + |5.2| = 10.9 \text{ meters}$
 - c. Determine the perimeter of the fence around the yard.
53.8 meters
 $2(16) + 2(10.9) = 32 + 21.8 = 53.8 \text{ meters}$



Prompt students to label the coordinates of the second fence on the graph to help determine the length and width. Consider having some students draw the fence on a piece of graph paper. The visual representation may be helpful as students reason about how to find the perimeter.

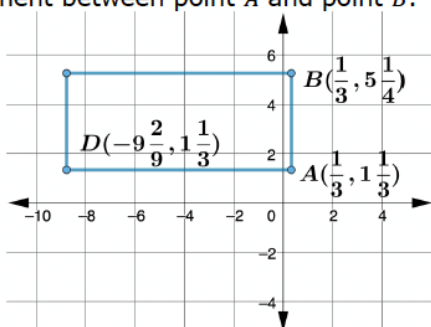
4.5.4 Collaboration: Fix the Mistake

Component Overview: Students engage in a *Fix the Mistake* activity where they look at two different students' work and answers for finding distance and perimeter on the coordinate plane. Students work collaboratively to determine the mistakes and correct them.



4.5.4 Collaboration: Fix the Mistake

- Vera and Conor are working on their homework together to find the perimeter of the rectangle, shown on the coordinate plane. The width of the rectangle is the segment between point A and point B .



Conor counts the units between the numbers, while Vera determines the dimensions by subtracting. Each student's work and answers are shown.

Vera's Thinking	Conor's Thinking
Vera finds the dimension of the width by subtracting the y-coordinates. $5\frac{1}{4} - 1\frac{1}{3} =$ $\frac{21}{4} - \frac{4}{3} =$ $\frac{63}{12} - \frac{16}{12} =$ $\frac{47}{12}$ Vera says the width of the rectangle is $3\frac{11}{12}$ units.	Conor notices there is 1 full unit between point A and point B and two pieces that look like they would combine to make another whole unit. Conor says the width of the rectangle is 2 units.

Discuss each student's work with your partner. Explain any errors that either student made.

Answers will vary. Vera's thinking is correct. Conor forgot that the graph is scaled by a factor of 2 when counting the units.



Think-Pair-Share

Position students to engage in a Think-Pair-Share to discuss each student's work. Students analyze the work independently first and then share their thoughts with a partner before explaining their response in writing. Circulate as students share with a partner and choose one to two students to share with the class the errors they found.

4.5.4 Collaboration: Fix the Mistake (continued)

2. The length of the rectangle is the segment between point A and point D .

Vera finds the dimensions of the length by subtracting the x -coordinates.

$$9\frac{2}{9} - \frac{1}{3} =$$

$$\frac{83}{9} - \frac{3}{9} =$$

$$\frac{80}{9}$$

Vera finds the length of the rectangle to be $8\frac{8}{9}$ units.

Conor counts the units between the numbers again. He counts 4 units between A and D , but then realizes each space on the grid represents 2 units, so he doubles this to 8. He then determines $\frac{2}{9}$ of a unit plus $\frac{1}{3}$ of another unit equals $\frac{5}{9}$. He adds this to the 8.

Conor finds the length of the rectangle to be $8\frac{5}{9}$ units.

Discuss each student's work with your partner. Explain any errors that either student made.

Answers will vary. Conor and Vera both made mistakes in their work. Vera should have added the absolute value of the x -coordinates instead of subtracting them since points A and D are on opposite sides of the y -axis.

Conor made a mistake with the scale again. Although he accounted for the scale when counting the whole units, he did not use the scale when counting the fractional units. He should have added $1\frac{5}{9}$ to 8 to get $9\frac{5}{9}$.

3. After collaborating with each other and understanding their mistakes, Vera and Conor each correctly find the perimeter of the rectangle. What perimeter did they find?

$$\text{Length of rectangle} = 9\frac{5}{9}$$

$$\text{Width of rectangle} = 3\frac{11}{12}$$

$$\text{Perimeter} = 2\left(9\frac{5}{9}\right) + 2\left(3\frac{11}{12}\right) = 26\frac{17}{18} \text{ units}$$



If students are struggling to determine the errors in the students' work, then consider the following prompts:

- What is the scale of the x - and y -axes?
- Why is this important in finding the distance between the points?
- How can you find the distance between points on the coordinate plane?
- How does the location of the points impact how you find the distance between them?
- Does it matter which side of the axes the points are located on? Why?



Consider splitting students into groups and having one group focus on Vera's work and the other on Conor's work. Have the groups be responsible for sharing out what they noticed, and then work together to fix the mistakes. Consider allowing students to explain verbally instead of writing.

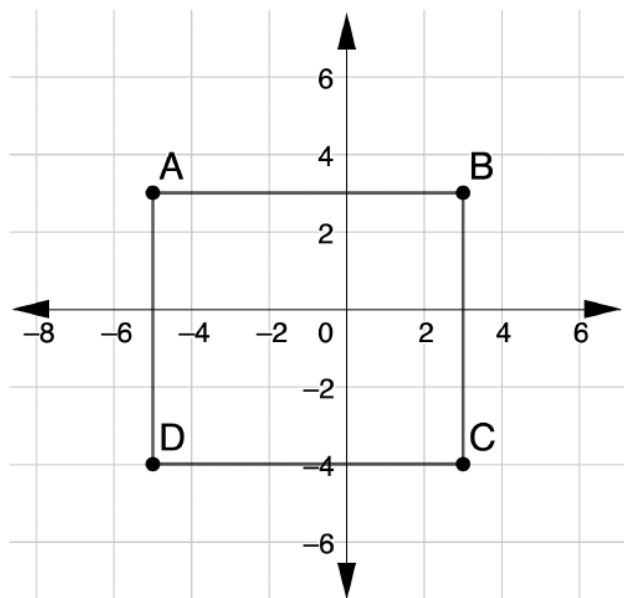
4.5.5 Your Turn!

Component Overview: Students find dimensions of a rectangle on the coordinate plane and use them to find the perimeter of the rectangle.



4.5.5 Your Turn!

1. The vertices of a rectangle are at points $A(-5, 3)$, $B(3, 3)$, $C(3, -4)$, and $D(-5, -4)$.



- What is the length, in units, of the rectangle?
8 units
- What is the width, in units, of the rectangle?
7 units
- What is the perimeter, in units, of the rectangle?
 $2(8) + 2(7) = 16 + 14 = 30$ units



Students may still struggle to accurately find the length and width of the rectangle. Encourage students to note the scale of the x - and y -axis. If students still struggle, provide additional scaffolding:

- Are the points on the same side of the axis or on opposite sides?
- How far are the points from the axis?
- How can we use this to help us find out how far the points are from each other?
- Does your answer make sense?
- How can you use the location and relation of the points on the coordinate plane to support your answer?

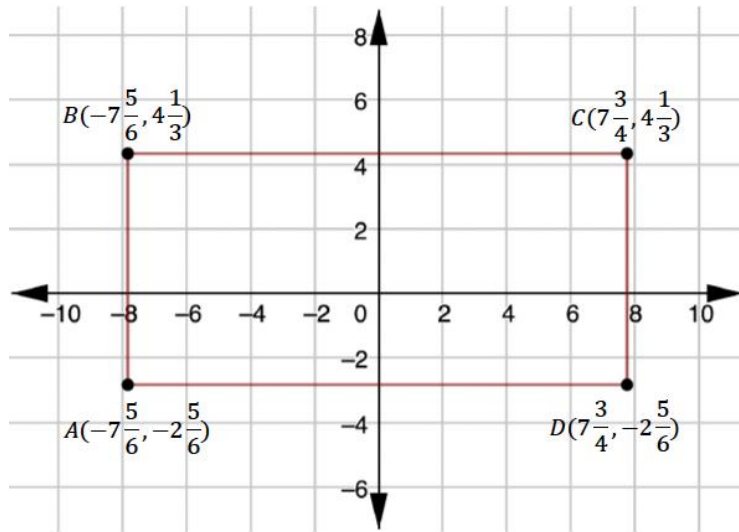


Notice the similarity between this question and the 4.5.6 Wrap-Up, particularly with the open-ended possibility of choosing which side is the length. Consider asking the following questions to the whole group:

- Which side is the length? How do you know?
- Is length always the longest side?
- If the length and width were switched, would the perimeter still be the same?

4.5.5 Your Turn! (continued)

2. Titus wants to put a border around the outside of his garden.



If the scale of the grid is in feet (ft), how many feet of border will he need to buy to completely fence in the garden?

Length of fence (A to B)

$$\left|4\frac{1}{3}\right| + \left|-2\frac{5}{6}\right| = 7\frac{1}{6} \text{ ft}$$

Width of fence (A to D)

$$\left|-7\frac{5}{6}\right| + \left|7\frac{3}{4}\right| = 15\frac{7}{12} \text{ ft}$$

Perimeter of fence

$$2\left(7\frac{1}{6}\right) + 2\left(15\frac{7}{12}\right) = 45\frac{3}{6} \text{ or } 45\frac{1}{2} \text{ ft}$$



Notice that students need to add fractions with different denominators in question 2. Consider substituting these fractions with friendlier numbers, if necessary.

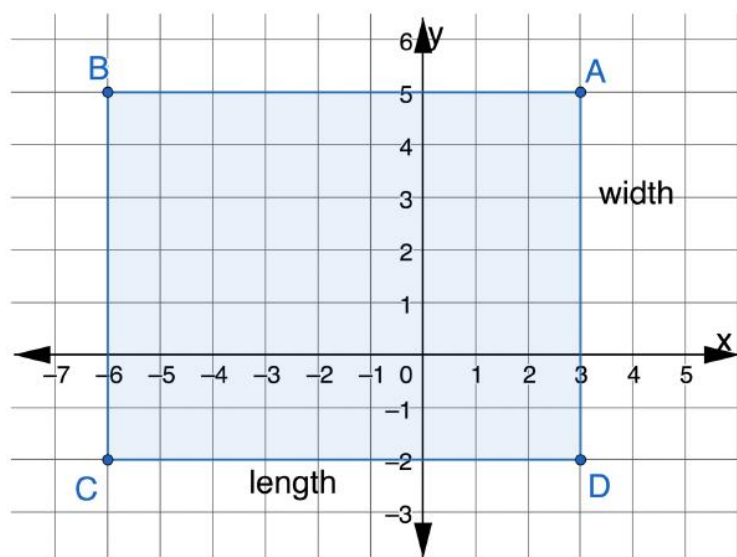
4.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students apply what they have learned to show understanding of finding perimeter of a rectangle on the coordinate plane.



4.5.6 Wrap-Up: Ticket Out the Door

1. A rectangle is shown on the coordinate plane.



Consider asking students to show their work or explain how they found their answer. This information will help inform you as to which students rely on counting and which students use absolute value.

- a. What is the length, in units, of the rectangle?
9 units
- b. What is the width, in units, of the rectangle?
7 units
- c. What is the perimeter, in units, of the rectangle?
 $2(9) + 2(7) = 18 + 14 = 32$ units

4.6 Solving Problems Involving Rectangles on the Coordinate Plane

Lesson Introduction

 Benchmarks	MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane. MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.		 Learning Targets	- I can find distances between ordered pairs. - I can find the area of a rectangle on a coordinate plane. - I can find the fourth vertex of a rectangle on a coordinate plane.
 Benchmark Coherence	Building On MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.		Working Toward MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.	
 Essential Questions	- How can you find the area of a rectangle on the coordinate plane? - How can you find the missing vertex of a rectangle on the coordinate plane?			
 Lesson Narrative	In this lesson, students will use their knowledge of the coordinate plane to help them find the area of rectangles. They will also use this knowledge to find the missing vertex of a rectangle on the coordinate plane. Students should use their prior knowledge from the previous lessons on distance on the coordinate plane, as well as what we know about the properties of a rectangle, throughout this lesson.			
 Component Summary	4.6.1 Warm-Up (~5 min.)	Complete a multiplication equation to make it true (<i>SE p. 166</i>)	4.6.4 Collaboration (10 min.)	Discuss the attributes of a rectangle and how they relate to the coordinate plane (<i>SE p. 168</i>)
	4.6.2 Collaboration (5-10 min.)	Find the area of a rectangle on the coordinate plane (<i>SE p. 166</i>)	4.6.5 Guided Practice (5-10 min.)	Determine the missing vertex of a rectangle (<i>SE p. 169</i>)
	4.6.3 Guided Practice (5-10 min.)	Find the area of rectangles on the coordinate plane (<i>SE p. 167</i>)	4.6.6 Your Turn! (5-10 min.)	Determine the missing vertex and area of a rectangle (<i>SE p. 169</i>)
	4.6.7 Wrap-Up (~5 min.)	Reflection on potential learning gaps (<i>SE p. 170</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Vertex - Parallel - Area	(Optional) Colored pencils (Optional) Graph paper	N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students might confuse x- and y-values in the coordinate plane. - Students may confuse when to add or when to subtract when finding distance on the coordinate plane.	- Students may wonder which side to call length and which side to call width. If these are not noted, then it should not matter which is named. Sometimes a general rule, such as length is the longer side and width is the shorter side is used in a classroom, but if this is done, it should be noted as a matter of convenience and not as a rule. However, the goal of the lesson is for students to find the lengths of the sides and use them to find area. At this point, it is not important which side they refer to as width and which they refer to as length. - Consider providing sentence frames for students to explain their reasoning.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 4.6.4 Collaboration, students engage in a discussion about the attributes of a rectangle on the coordinate plane. This discussion could be used as a Notice and Wonder to position students to think about the x -coordinates of the points that create a vertical line and the y -coordinates of the points that make a horizontal line and how they relate.	MA.K12.MTR.5.1: Patterns and Structure Students should relate what they already know about rectangles, i.e., what makes a shape a rectangle, finding area, etc., and apply that knowledge to finding the missing vertex and area of a rectangle on the coordinate plane.
 Differentiate Instruction	This lesson provides multiple visual entry points through graphs on the coordinate plane. Students use the grid to find missing vertices in the rectangles and can also use the visual of the rectangle to reason about the area. Auditory entry points are provided through collaborative activities and partner discussion. Consider how best to pair students for each Collaboration (4.6.2 and 4.6.4) to ensure discussions are fruitful and impactful to understanding. Data collected in the previous lesson can be used to inform partner pairs or small groups.	
Lesson Notes:		

4.6.1 Warm-Up: True Statement

Component Overview: Students fill in the boxes to make the multiplication equation true, using digits one through nine.



4.6.1 Warm-Up: True Statement

1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\boxed{1} \boxed{5} \times 3.2 = \boxed{4} \boxed{8}$$



Consider using this Warm-Up as an opportunity for informal observation of student proficiency in multiplying decimals. Students will be finding area using rational numbers throughout this lesson, and informal data from this Warm-Up can inform partner pairs or small groups for the 4.6.2 and 4.6.4 Collaboration activities later on.

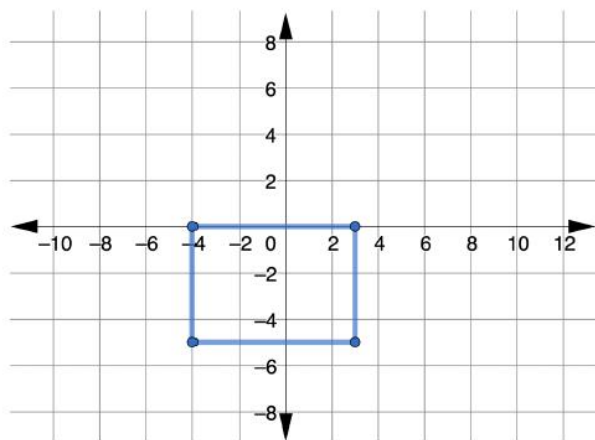
4.6.2 Collaboration: Area of Rectangles

Component Overview: Students collaborate with a partner to review the formula for area of a rectangle. Students then use the formula to find the area of a rectangle on the coordinate grid.



4.6.2 Collaboration: Area of Rectangles

1. Work with your partner to explain how to find the area of a rectangle.
Area of a rectangle = Length $\times$ Width
2. A rectangle with coordinate $(3, 0)$, $(3, -5)$, $(-4, 0)$, and $(-4, -5)$ is shown. The scale of the grid is in centimeters (cm).



Work with your partner to answer the following questions.

- a. Determine the length and the width of the rectangle. Include units.
Length = 5 cm
Width = 7 cm
- b. What is the area of the rectangle? Include units.
 $A = 7 \times 5 = 35 \text{ cm}^2$



Consider highlighting student responses that represent finding area as an equation, to encourage mathematical fluency as students answer questions with area throughout this lesson.

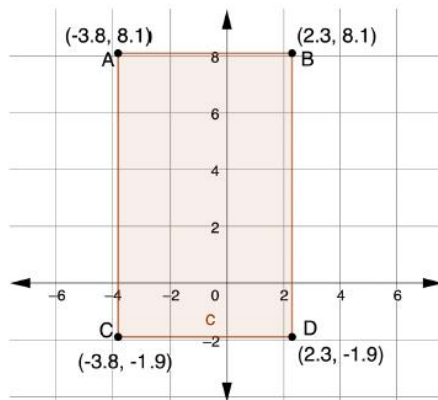
4.6.3 Guided Practice: Finding the Area of Rectangles on the Coordinate Plane

Component Overview: Students find the areas of rectangles on the coordinate plane. Encourage students to apply what they have learned in the last few lessons on finding distance between points on the coordinate plane, as well as finding perimeters of rectangles on the coordinate plane.



4.6.3 Guided Practice: Finding the Area of Rectangles in the Coordinate Plane

- Henry wants to install new carpet in his living room. A drawing of his living room, with measurements in feet, is shown.



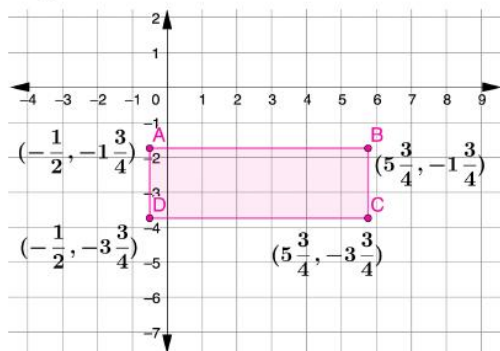
What is the area of the living room?

Length of room = 10 ft

Width of room = 6.1 ft

Area = $6.1 \times 10 = 61 \text{ ft}^2$

- A rectangle is shown.



What is the area of the rectangle?

Length of rectangle = 2 units

Width of rectangle = $6\frac{1}{4}$ units

Area of rectangle = $6\frac{1}{4} \times 2 = 12\frac{1}{2} \text{ units}^2$



Florida's assessment limits for MA.6.GR.1.2 and MA.6.GR.1.3 state that all ordered pairs should be integers. The rational ordered pairs in 4.6.3 and 4.6.5 help to support the area of emphasis, performing all four operations with integers, positive decimals, and positive fractions with procedural fluency, while also continuing student understanding of finding distance and area. By stretching a student's understanding beyond integers, the student will not necessarily be able to rely on counting to determine the distance, thus solidifying his/her understanding of absolute value as the distance from zero.



Consider offering graph paper for students to draw and label the rectangle with its dimensions after the length and width are found. Students may be more hesitant to engage and make mistakes if they only have one coordinate plane to use.



Encourage students to use the strategies they practiced in the previous lessons to determine length and width before finding the area. Draw attention to the scale and ensure students see that the points do not fall directly on the lines. This may be difficult for some students to see because the points are so close to the lines.



For some students, it may be helpful to substitute the coordinates with friendlier integers to allow them to focus on the process of finding distance and area rather than operations with fractions. This will provide an additional entry point without deterring from the learning goals of the lesson.

4.6.4 Collaboration: Finding the Fourth Vertex

Component Overview: Students collaborate to discuss all the attributes they know about rectangles. Prompt students to think about things like side length. Students will apply these ideas to determine where the missing vertex of a rectangle should be placed.



4.6.4 Collaboration: Finding the Fourth Vertex

- With your partner, list every attribute you know about rectangles.

Rectangles

Answers will vary.

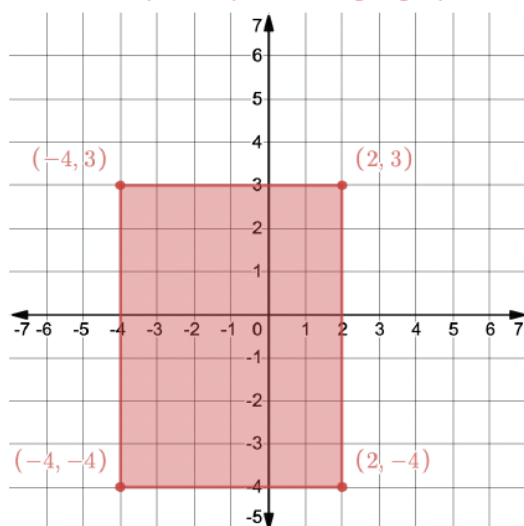
I know that rectangles have two sets of parallel sides.

I know that rectangles have two sets of equal sides.

I know rectangles have four right angles.

- With your partner, find another pair of partners. Share what you wrote about rectangles. Add any new attributes the other group found that you didn't.
- On the coordinate grid, draw a rectangle whose sides lie on the gridlines. Label the ordered pair of each vertex.

Answers will vary. Sample rectangle graphed below.



Highlight student responses that include vocabulary like:

- Parallel
- Equivalent/equal
- Side length
- 90 degrees



Before students begin, consider reviewing the definition of attributes to ensure all students understand the prompt. It may be particularly helpful to review attributes of something students are already familiar with, such as a classroom object, to provide additional examples and nonexamples of attributes.

4.6.4 Collaboration: Finding the Fourth Vertex (continued)

4. Discuss with your partner how to determine where to place the vertices of your rectangle. Write a summary of your discussion.

Answers will vary.

We knew that the horizontal lines both needed to have the same y -coordinates, and the vertical lines both needed to have the same x -coordinates.

5. Based on the rectangle drawn in the previous questions, add any attributes about rectangles to your list.

Answers will vary.



For students who finish early, consider an extension in which students find the area and perimeter of the rectangle they drew.

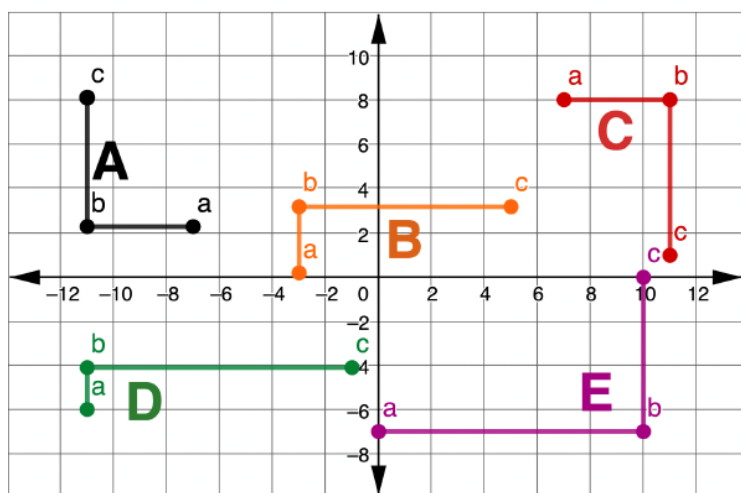
4.6.5 Guided Practice: Finding the Fourth Vertex

Component Overview: Students find the missing vertex in a rectangle on the coordinate plane. Students apply what they know about the attributes of a rectangle as discussed in the 4.6.4 Collaboration to finish each rectangle on the grid.



4.6.5 Guided Practice: Finding the Fourth Vertex

- Using the coordinate plane, find the fourth vertex of the following rectangles.



	Vertex a	Vertex b	Vertex c	Vertex d
Rectangle A	(-7, 2.3)	(-11, 2.3)	(-11, 8.1)	(-7, 8.1)
Rectangle B	(-3, 0.2)	(-3, 3.2)	(5, 3.2)	(5, 0.2)
Rectangle C	(7, 8)	(11, 8)	(11, 1)	(7, 1)
Rectangle D	(-11, -6)	(-11, -4.1)	(-1, -4.1)	(-1, -6)
Rectangle E	(0, -7)	(10, -7)	(10, 0)	(0, 0)

- What is the area of Rectangle A, in square units?

$$\text{Area of rectangle} = 4 \times 5.8 = 23.2 \text{ units}^2$$



Encourage students to apply what they know about the attributes of rectangles to determine the missing points.

- If the sets of lines must be parallel, where can you place the final vertex to create a parallel line?
- How can you use the coordinates of each point to determine the coordinates of the fourth point?
- Where do you need to place the vertex to create a right angle? Why?

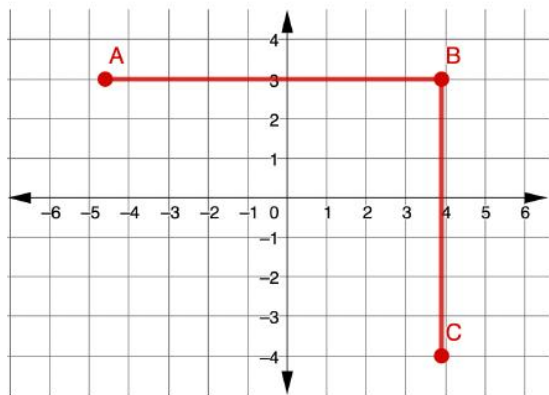
4.6.6 Your Turn!

Component Overview: Students find the missing vertices of the rectangles and then determine the area of each rectangle.



4.6.6 Your Turn!

1. Three vertices of a rectangle are at points $A(-4\frac{3}{5}, 3)$, $B(3\frac{9}{10}, 3)$, and $C(3\frac{9}{10}, -4)$. Each gridline represents 1 centimeter.



- a. What are the coordinates of the fourth vertex?

$$(-4\frac{3}{5}, -4)$$

- b. What is the area of the rectangle?

$$\text{Length of rectangle} = 7 \text{ cm}$$

$$\text{Width of rectangle} = 8\frac{1}{2} \text{ cm}$$

$$\text{Area of rectangle} = 7 \times 8\frac{1}{2} = 59.5 \text{ cm}^2$$

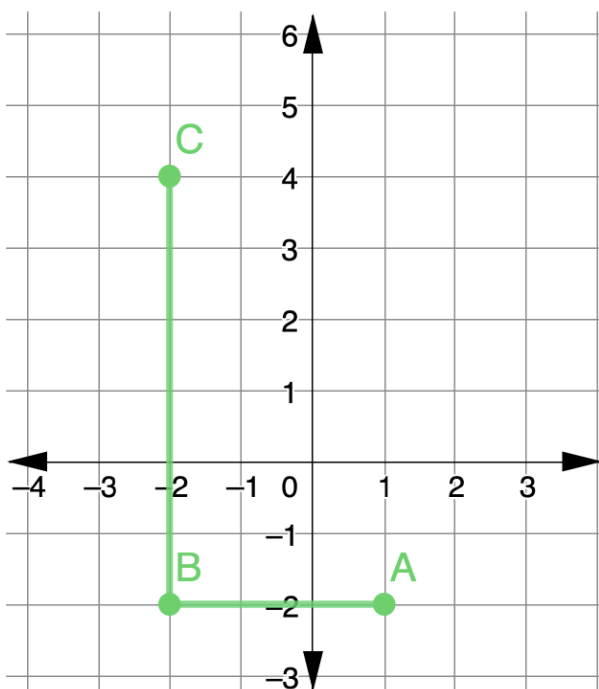


Consider challenging early finishers with the following questions:

- How could you justify that the shape is not a square?
- How would the vertices need to change for the shape to be a square?

4.6.6 Your Turn! (continued)

2. Jada needs to cut out a piece of fabric in the shape of a rectangle. She laid out the fabric on a cutting board, whose scale is represented in feet (ft). Jada placed two edges of the fabric at points A and B . She wants the length of the fabric to be as long as the distance from point B to point C .



- a. Determine the coordinates of the fourth corner of the fabric.
The fourth coordinate would need to be on the point (1,4).
- b. Find the area of the piece of fabric.
Length of fabric = 3 ft
Width of fabric = 6 ft
Area of fabric = $3 \times 6 = 18 \text{ ft}^2$



Encourage students to use the graph and ask themselves “Does my answer make sense?” When applicable, students can count the boxes inside the graph to check their answer for area.

4.6.7 Wrap-Up: Reflection

Component Overview: Use the Wrap-Up to help determine which topics should be reviewed prior to the unit assessment.



4.6.7 Wrap-Up: Reflection

1. Complete the following statements.
 - a. Today, I am still unsure about ...
Answers will vary.
 - b. To fill this gap, I intend to ...
Answers will vary.



Use the 4.6.7 Wrap-Up to help determine which topics may need remediation prior to the unit assessment.